

Semantics of obviation in Atchan: Disjointness and long-distance obligatory binding*

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Abstract This paper documents and analyzes a contrast between two 3SG pronouns, the proximate and the obviative, in Atchan (Kwa, Côte d’Ivoire). I show that the obviative has a restricted distribution and restricted referential possibilities, both of which can be captured on a treatment of this pronoun as an obligatorily bound element. I develop a semantic account of this pronoun’s distinctive behaviors on which the obviative contains a presuppositional disjointness feature. I show that this account captures the anti-prominent orientation and restricted distribution of the obviative.

Keywords: binding, pronouns, obviation, logophoricity, reference tracking, African languages

1 Introduction

In many cases, it is not immediately obvious which module(s) of the grammar should be responsible for capturing licensing phenomena. Specifically, the division of labor in licensing between the syntax and the semantics is often unclear. In the domain of polarity, for example, we have seen an explanatory shift in which earlier syntactic explanations of NPI licensing (e.g., Klima 1964) have given way to more semantic accounts like those based on domain widening (Kadmon & Landman 1993, Krifka 1995, Chierchia 2013, a.m.o.). A similar question concerns the

* *Nansio* to Jeanne Doko, Léocadie Guebeapo, Chrystelle Gomon, Evelyne Koutouan, Honoré Koutouan, and Marcel Paul for sharing their language with me. I especially thank Maxime Dido for discussion of some of the data reported here. The ideas in this paper have been advanced by conversation with and questions from Chris Barker, David Blunier, Lucas Champollion, Isabelle Char-navel, Cleo Condoravdi, Amy Rose Deal, Peter Grishin, Vera Hohaus, Peter Jenks, Line Mikkelsen, Will Oxford, Justin Royer, Katherine Russell, Malte Zimmermann, and audiences at Stanford, UC Berkeley, TripleA 11, Sinn und Bedeutung 29, the Paris Workshop on Varieties of Perspective Shift, and the McGill Understanding Obviation workshop. I additionally thank Florian Schwarz and three anonymous reviewers at NLS for their detailed and thoughtful comments and suggestions. This field-work was financially supported by Oswald Endangered Language Grants from UC Berkeley and a Lewis and Clark Fund for Exploration and Field Research Grant from the American Philosophical Society. All remaining errors are my own.

licensing of locally-bound anaphors, which have been understood either to be subject to a syntactic requirement (Chomsky 1981, Charnavel & Sportiche 2016) or to involve a semantic reflexivization operation (Bach & Partee 1980, Pollard & Sag 1992, Reinhart & Reuland 1993, Sauerland 2013, McKillen 2016).

In this paper, I raise this question in another domain using a case study of a pronominal contrast in Atchan, a Kwa language of Côte d'Ivoire. Atchan exhibits two 3SG animate pronouns, which, inspired by the literature on Algonquian obviation systems (Bloomfield 1946, Rhodes 1990, Aissen 1997, Thomason 2003, a.m.o.), I gloss as PROX 'proximate' and OBV 'obviative'.¹ Notably, these pronouns exhibit different coreferential possibilities:²

- (1) a. timote_j pɔ mɛ̃_{i/j} wó
 T. love PROX cat
 'Timothée_j likes 3SG_{i/j}'s cat.'
- b. timote_j pɔ nɛ̃_{i/*j} wó
 T. love OBV cat
 'Timothée_j likes 3SG_{i/*j}'s cat.' (20220805_chr_mar)

In (1a), PROX can refer to the subject Timothée or have disjoint reference; OBV, by contrast, must be disjoint from the subject in (1b).

This paper has two main goals. The first is to argue that the alternation between PROX and OBV should be viewed as a phenomenon that involves licensing. Specifically, as I will show, OBV cannot freely occur in all positions in all sentences. Instead, both its distribution and its referential possibilities are restricted in a way that can be cleanly captured by the generalization that OBV is obligatorily bound, i.e., requires a particular kind of licensing. Of course, however, the striking thing about OBV in (1b) is that it must be *disjoint*; this contrasts with the obligatory *identity* effects that arise with better-known phenomena like Condition A-obeying anaphors. My claim about obligatory binding in Atchan, therefore, includes the claim that obligatory binding can manifest in the form of obligatory disjointness effects.

The second goal of this paper is to develop a view on which this obligatory binding of OBV is enforced in the semantics. Inspired by Koopman & Sportiche (1989), who analyze similar disjointness effects in the Kwa language Abé as involv-

1 This terminological choice is intended to parallel the literature on Algonquian obviation. Algonquian obviation, and the Atchan system discussed here, differ from another phenomenon in Romance and other European languages also termed 'obviation', which involves different coreference patterns of subjects in subjunctive clausal complements. See §7.1 for a brief discussion of Romance 'obviation'. Note also that, on the analysis I pursue, the 'proximate' pronoun is in all respects an ordinary 3SG pronoun.

2 Throughout this paper, I indicate the availability or unavailability of particular coreferential/binding possibilities via subscripts in the translation. Unless explicitly noted otherwise, my claim is merely that the indicated coindexation or binding possibility is available (or is not, as the case may be).

ing binding, and [Bassi et al. \(2023\)](#), who develop an identity-based presuppositional account of logophoric pronouns, I develop an account of OBV on which it introduces a disjointness presupposition, adding a second variable that is bound by the highest nominal in a specific portion of the matrix-clause structure. The combination of a non-identity presupposition and binding of a variable in that presupposition yields disjointness effects through the machinery of semantic binding. Taking up the question of where the explanatory weight of OBV’s behavior lies, I also show that this phenomenon does not display the locality effects that we expect of purely syntactic phenomena. Instead, I argue that the unbounded, truly long-distance nature of the relationship between OBV and its licenser speaks in favor of a semantic account.

The paper proceeds as follows. I provide a descriptive introduction to Atchan’s pronominal system in §2. In §3 and §4, I develop two empirical generalizations about the restricted distribution and referential possibilities of OBV, which together lay the groundwork for the claim that OBV is subject to an obligatory-binding constraint. In §5 and §6, I develop a semantic account of OBV’s behavior. In §7, I show that OBV does not behave as expected on a range of purely syntactic approaches to long-distance pronominal licensing. Finally, I situate Atchan in a larger landscape of reference-tracking phenomena in §8 and conclude with §9.

2 Atchan and its pronouns

2.1 Language and consultant backgrounds

Atchan (pronounced [ácâ], also called Ébrié; ISO: ebr; SVO) is a Kwa (Niger-Congo) language spoken by the Tchaman people in villages within and around the city of Abidjan, Côte d’Ivoire. Due to the geographic location of the Tchaman villages within Abidjan, Côte d’Ivoire’s economic capital, the Tchaman people have been in long-term sustained contact with French; nearly all Atchan speakers, including all of my consultants, are bilingual in Atchan and French (and, often, additional other Ivorian languages).

The novel data discussed in this paper comes from two sources. One source is elicitation with six main Atchan speakers between 2021-2025, in a combination of in-situ and remote work. These speakers live in two villages, Anono and Blockhauss, which are located in the city of Abidjan. All data from elicitation is reported with a code corresponding to the date of the relevant elicitation session and the speaker(s) who participated in that session. Recordings and notes from the Atchan Language Project, including all elicitation files, are archived with the California Language Archive ([Doko et al. 2021](#)).

The second source of data discussed in this paper comes from the translation of the New Testament into Atchan ([Loba & Biekre 1997](#)), which was conducted by

	S form	O form	elsewhere
1SG	mẽ		
2SG	ε	hε	
(3SG.)PROX	ã/ẽ~ń	-ε	mẽ
(3SG.)OBV	nε~nkε		
1PL	lo		
2PL	ǔ	hǔ	
3PL	wo		
3INAN	á	∅	ló

Table 1 Personal pronouns in Atchan, adapted and updated from Bôle-Richard (1983).

two Atchan native speakers (Janvier Blewoué, p.c.). At present, the New Testament translation is the only publicly-available large text in Atchan. I have confirmed key portions of the text with my consultants, and I additionally have confirmed that central generalizations from elicitation regarding the distributions of Atchan pronouns are upheld in the text. The text exhibits certain dialect-related differences from the variety of Atchan that my consultants use (e.g., in the forms of the pronouns and orthographical word conventions); those differences are discussed as relevant. New Testament examples are provided with my own literal translations, with reference to the relevant chapter and verse(s).

This paper uses slightly different orthographical conventions for examples from different sources. All examples from elicitation sessions are transcribed in IPA, with H tones indicated with acute accents and L tones unmarked. Examples cited from Dido's (2018a) grammar of Atchan follow Dido's transcriptions, though I have standardized tone marking to only indicate H tones and adapted glosses. The New Testament translation uses a different orthography, which roughly aligns with the Practical Orthography for Ivorian Languages (Kokora et al. 1996); these examples are presented with the translation orthography followed by an IPA transcription.

Throughout the paper, glossing follows the Leipzig Glossing Rules, with the following additions: CM class marker prefix, O object, OBV (3SG) obviative, PART verbal particle, S subject.

2.2 Atchan pronouns

Our discussion of Atchan in this paper is centered around its pronouns, so it is imperative that we understand the pronominal inventory of Atchan. In this section, I provide a descriptive overview of the personal pronouns in Atchan; the full inven-

tory of personal pronouns used by my consultants is shown in Table 1. In this paper, we are primarily interested in the two third-person singular animate pronouns, which I gloss PROX ‘proximate’ and OBV ‘obviative.’³ Note that both PROX and OBV can only refer to animate entities; I will not focus on the animacy restriction in this paper.

My claim that Atchan has these two distinct third-person singular animate pronouns is a novel descriptive claim; earlier literature (Bôle-Richard 1983, Dido 2018a) conflated the two into a single 3SG pronoun. For this reason, it is worth walking through in detail both the different forms of the two pronouns and the evidence that PROX and OBV are indeed separate pronouns.

First, let us focus our attention on PROX. The form of this pronoun depends on three factors: structural position, aspect, and dialect. First, PROX has distinct subject ([ã]/[é]/[ń]), object ([-ε]), and elsewhere ([m̃]) forms. The PROX.O form, [-ε], phonologically coalesces with the verb; I supply the underlying form of the verb for every coalesced example of PROX.O we see. Second, the form of PROX.S varies based on the clause’s aspect. As is areally common among West African languages (Anderson 2011, 2015, 2017, Garvin et al. To appear), Atchan expones aspect distinctions on subject pronouns; [ã] PROX.S is non-perfective, while [é]/[ń] PROX.S.PFV is perfective. Finally, the distinction between the two PROX.S.PFV forms [é] and [ń] reflects dialect variation; my consultants use [é], while the Atchan New Testament and Dido’s thesis use [ń].

I claim that Atchan has a second 3SG pronoun, [n(k)ε] OBV, whose form does not vary by structural position or aspect. Its form does show dialect-based variation; the consultants that I work with mostly use [nε], while the Atchan New Testament uniformly uses [nkε] (and, in his thesis, Dido (2018a) writes this form as [ŋkε]). This form was previously analyzed as a second object form, in free variation with [-ε] (Bôle-Richard 1983 p. 345, Dido 2018a p. 81). However, Dido (2018a,b) also notes that [n(k)ε] can occur in non-object positions, complicating this generalization.

Instead, I argue that [n(k)ε] OBV is a fully-distinct pronoun. The cleanest evidence for the distinctness of PROX and OBV comes from textual data. As will be discussed further in §4.2, the distinction between PROX and OBV is leveraged in texts like the Atchan New Testament translation (Loba & Biekre 1997) for refer-

3 Bôle-Richard (1983) and Dido (2018a) mention that some speakers contrast two 3PL forms [wo] and [nkõ]. I suspect that, for those speakers, [nkõ] is a plural obviative form. However, none of my consultants use [nkõ], so this paper focuses solely on the singular forms.

ence tracking. An example passage, with PROX referring throughout to one individual and OBV to another, is given below:⁴

- (2) Erɔde esó Jan. Nýen salé nke li lephan bhwá, nke te créncre, lóhéthé áno
 nke mwámwá. Lóka anyé nke bhwe nke tɔ, álri mén thén gbre, lóte khén
 nke bhwetó limén yé.
 ɛrɔde_i e-só jã_j. ní-pẽ salé nke_j li lep^{hã}
 Herod PROG-fear John PROX.S.PFV-know COMP OBV COP person
 bhwá, nke_j te créncre, ló-hént^hé á_i-mbo nke_j
 righteous OBV COP holy 3INAN-because.of PROX.S-protect OBV
 mwámwá. ló-ka ã_i-pé nke_j bwe nke_j tɔ, á-lri
 good DEF-time PROX.S-hear OBV speech OBV say 3INAN.S-bother
 mé_i th^é gbre, ló-te k^hé nke_j bwetó limé_i jé.
 PROX heart bother DEF-? COMP OBV speech please.PROX.O hear
 ‘Herod_i feared John_j. He_i knew that he_j was a righteous person, (that) he_j
 was holy; because of this, he_i was protecting him_j well. When he_i heard the
 speech that he_j said, it bothered his_i heart, but his_j speech pleased him_i.’
 (Mark 6:20)

In this passage, Herod is referred to using forms of PROX ([ã]/[ń] PROX.S/PROX.S.PFV, [-ɛ] PROX.O,⁵ and [mé] elsewhere PROX), while John is uniformly referred to with OBV (invariant [nke]).

The systematicity of alternations like this one suggests that PROX and OBV are indeed two separate pronouns. If they were one pronoun, and [n(k)ɛ] and other forms freely alternated with one another, it would be hard to explain how these systematic reference-tracking effects could obtain. In the rest of this paper, our focus is on the meanings of Atchan PROX and OBV, and how effects like these reference tracking ones can be derived.

In the next two sections, I discuss the restricted distribution of OBV and the disjoint reference effects that it exhibits. I argue in support of the following two empirical generalizations about OBV, which I term its *distributional constraint* and its *reference constraint*:

- (3) **Distributional constraint on OBV:** Atchan OBV cannot occur as the highest nominal in the matrix Top > Foc > Subj sequence.
 (4) **Reference constraint on OBV:** Atchan OBV cannot be construed as coreferential with or bound by the highest nominal in the matrix Top > Foc > Subj sequence.

⁴ In this passage, Herod has conflicting feelings about John. The New International Version translation of Mark 6:20 is ‘Herod_i feared John_j and protected him_j, knowing him_j to be a righteous and holy man. When Herod_i heard John_j, he_i was greatly puzzled; yet he_i liked to listen to him_j.’

⁵ In this passage, we see PROX.O in the form [limé]. The underlying form is /limã-ɛ/ ‘please-PROX.O’.

The next section, §3, provides evidence to support the distributional constraint. Afterwards, §4 focuses on the reference constraint.

3 The distributional constraint

In this section, I argue in support of the constraint in (3). My evidence for this constraint comes from three sources. First, OBV resists occurring in matrix subject, focus, and topic positions. Second, this is a sentence-level constraint: OBV’s resistance to occurring in these high positions disappears in embedded clauses. Finally, in the presence of higher left-peripheral nominals, OBV can occur in comparatively lower positions in this sequence.

In Atchan, the matrix-clause periphery is structurally complex. Topics linearly precede (and, I assume, c-command) focused elements, which in turn precede subjects:

- (5) Mɛn-ɛ, ɲɔdu nɔn mɛn mo mɛn há hɔn hɛn
 [top. mɛ-ɛ,] [foc. ɲ-du nɔ] [subj. mɛ] mo mɛ há hɔ hɛ
 1SG-TOP CM-water FOC 1SG take 1SG pour 2PL head
 ‘Me, it’s water that I take and pour on your heads.’ (Matthew 3:11)

Matrix topics are always set off by a prosodic break (indicated with a comma) from other material in the sentence. The TOP suffix overtly marks matrix topics in New Testament translation examples and in most elicitation examples with H-toned topics. The focus construction in Atchan involves the marker *nɔ* FOC, which follows the focused element. Atchan is SVO, and subjects immediately precede the verb or any auxiliaries. In [Jarvis 2025a,b](#), I showed that matrix-clause topics are base-generated in the clausal periphery, while focused items A’-move to the periphery.

As illustrated below, each of these positions is one where OBV is dispreferred in isolation. In this data, we see that OBV is dispreferred in subject position in (6a), focus position in (6b), and topic position in (6c).

- (6) a. ɛ́/#nɛ c^ha ló t^ho
 PROX.S.PFV/OBV step 3INAN top
 ‘3SG stepped on it.’ (20230623_kou_jea)
 b. mɛ́/#nɛ nɔ mɛ mpɔ
 PROX/OBV FOC 1SG love
 ‘It’s 3SG that I like.’ (Maxime Dido p.c.)⁶
 answer to ‘Who do you like?’

⁶ Maxime Dido (p.c.) notes that it’s best to use PROX here because “[if OBV is used] the other person seems very confused if the context of communication is not known in advance by the interlocutors.”

- c. elize/#nɛ_i, mẽ nɔ̌ mẽ ndɔ mã nɛ_i mɔ̌
 L./OBV 1SG FOC 1SG help OBV PART
 ‘Lindsay/#3SG_i, it’s me who helped 3SG_i.’ (20230628_hre_chr_mar)⁷

The uniformity of this dispreference speaks in favor of a view on which OBV cannot occur as the highest nominal in the sentence.

A second piece of evidence that supports this generalization is that OBV does not resist *all* subject and focused occurrences. The evidence comes from clausal embedding. While topicalization does not obtain in embedded clauses, embedded clauses may show the same focus dislocation as matrix clauses, and of course can have subjects. In texts, OBV naturalistically occurs in embedded subject (7a) and focus (7b) positions:

- (7) a. íwu salé nɛ pɛ mímíbhú....
 í-_i-wu salé [nɛ_j pɛ mímíbhú...]
 PROX.S.PFV-see COMP OBV have faith
 ‘He_i saw that he_j was faithful...’ (Acts 14:9)
- b. Lóka wo yɛn salé nɛ nɔn nɛ li lephan...
 ló-ka wo pẽ salé [nɛ nɔ̌ nɛ li lep^hã...]
 DEF-time 3PL know COMP OBV FOC OBV COP person
 ‘When they realized that it was he who was the person...’ (Acts 3:10)

Instead, OBV resists occurring, specifically, in *matrix-clause* subject and focus position: this is consistent with our generalization, since embedded subject/focus is necessarily not the highest position in the sentence.

A final piece of evidence in support of this generalization comes from layering effects. Here, we focus directly on the Top > Foc > Subj structural hierarchy. As seen earlier, OBV generally resists occurring in matrix subject and focus positions. However, when subjects and focused items are combined with an overt matrix topic, OBV is exceptionally permitted in these positions:⁸

- (8) a. Setup (all in Atchan): Q: ‘Tell me something about Lindsay. Who chased her?’ A: ‘It’s me who chased.PROX.O.’ Q: ‘Now, tell me about Katie. Who chased her?’
 kati_j, nɛ_i nɔ̌ nɛ_i trẽmẽ_j
 K. OBV FOC OBV chase.PROX.O
 ‘Katie_j, it’s 3SG_i who chased 3SG_j.’ (20230803_hre)

⁷ The reader might be surprised by the fact that OBV can resume the matrix-clause topic here. We will return to this in §6.3.

⁸ The verb in (8a) is [trẽmẽ], underlyingly /trãmã-ɛ/ chase-PROX.O. The two verbs in (8b) are [krẽmẽ] (/krãmã-ɛ/ tell-PROX.O) and [pwɛ] (/pɔ-ɛ/ love-PROX.O)

- b. Setup (all in Atchan): Q: ‘Tell me something about Lindsay. What did she hear?’ A: ‘Lindsay, I told her that I like her.’ Q: ‘Tell me something about Katie. What did she hear?’

kati_j, nɛ_i krẽmẽ_j lé nɛ_i pwɛ_j
 K. OBV tell.PROX.O COMP OBV love.PROX.O
 ‘Katie_j, 3SG_i told 3SG_j that 3SG_i likes 3SG_j.’ (20230803_hre)

Here, (8a) involves the combination of a matrix topic and focus; we can see that focused OBV here is deemed acceptable. Similarly, in (8b) OBV is permitted as a (non-focused) subject in the presence of a matrix topic.⁹ This again supports our generalization: OBV can exceptionally occur in structurally high positions in the clause when even higher positions are occupied.

4 The reference constraint

In this section, I show that OBV is subject to disjoint reference effects that constrain its referential and binding possibilities. The constraint I argue for is given in (4), repeated below:

- (9) **Reference constraint on OBV:** Atchan OBV cannot be construed as coreferential with or bound by the highest nominal in the matrix Top > Foc > Subj sequence.

In §4.1, I show that OBV exhibits disjoint reference effects from structurally-high items, both locally and non-locally. In §4.2, I show that in information-structurally-rich contexts like narrative texts, the contrast between PROX and OBV gives rise to a full reference-tracking system, with OBV not referring to the topic.

4.1 Disjoint reference effects

The goal of this section is to show that OBV exhibits disjoint reference effects that require it to be disjoint from the highest Top > Foc > Subj nominal in the sentence, regardless of (a) how local the highest nominal and OBV are to each other and (b) what information-structural projection the highest nominal occupies. We also observe similar effects in the domains of pure coreference and quantificational binding.

As we saw at the beginning of the paper, the (co)referential possibilities of Atchan OBV are restricted relative to its counterpart PROX. This is shown again

⁹ Because focus in Atchan involves movement, testing the effect of overt focus on subject OBV is challenging: in addition to the inherent implausibility of such sentences, this configuration gives rise to a crossover configuration on the analysis pursued here.

below. In out-of-the-blue sentences, OBV cannot corefer with the matrix-clause subject, both when the two occupy the same clause (10b) and when OBV occurs in an embedded¹⁰ clause (11b). By contrast, PROX can optionally corefer with the subject in these examples:

- (10) a. *timote_j pɔ mɛ̃_{i/j} wɔ*
 T. love PROX cat
 ‘Timothée_j likes 3SG_{i/j}’s cat.’
 b. *timote_j pɔ nɛ_{i/*j} wɔ*
 T. love OBV cat
 ‘Timothée_j likes 3SG_{i/*j}’s cat.’ (20220805_chr_mar)
- (11) a. *timote_j ɓu sákolé [ã_{i/j} má má]*
 T. think COMP PROX.S FUT come
 ‘Timothée_j thinks that 3SG_{i/j} will come.’
 b. *timote_j ɓu sákolé [nɛ_{i/*j} ɓa ɓá]*
 T. think COMP OBV FUT come
 ‘Timothée_j thinks that 3SG_{i/*j} will come.’ (20220805_yap)

These examples show that, out of the blue, OBV cannot corefer with a matrix subject, and that this pattern holds across clause boundaries.

Note that, as is consistent with our generalization, OBV does not need to be globally free: OBV can corefer with a higher element if that element is relatively structurally low. Below, we see OBV coreferring with and being quantificationally bound by the addressee of *tell*:¹¹

- (12) a. *timote krāmā lízi_i lê nɛ_i ɓi ɓɔ*
 T. tell L. COMP OBV child pretty
 ‘Timothée told Lindsay_i that 3SG_i’s child is pretty. (20230629_yap)

10 Note that my consultants use different forms of the complementizer: one consultant (with code yap) tends to use *sákolé*, while other consultants often use *salé* or *lê*. Impressionistically, the form *lê* is also more common with speech verbs, though at least most verbs are compatible with multiple forms of the complementizer. To my knowledge, this interspeaker variation has no effect on the phenomena under study here.

11 All existing evidence at this point suggests that addressees of *tell* in Atchan c-command DPs in the clausal complement of *tell*. For example, a Condition C violation obtains between the addressee and R-expressions in the complement clause:

- (1) *mɛ̃ ŋkrāmā nɛ_{*i} lê kati_i ɓɔ*
 1 SG tell OBV COMP K. pretty
 ‘I told 3SG_{*i} that Katie_i is pretty.’ (20230706_yap)

- b. timote krāmā [lep^hã brɛbrɛ]_i lê nɛ_i fi pɔ̃
 T. tell person each COMP OBV child pretty
 ‘Timothée told each person_i that 3SG_i’s child is pretty.’
 (20230628_hre_chr_mar)

That is, the crucial element of the disjointness effects that we have seen so far in this section concerns OBV’s relationship to the matrix subject.

The disjoint reference effects are more nuanced than a simple ban on subject coreference, in two ways. First, the ban only involves *matrix* subjects, not all subjects; and second, elements occupying information-structural projections similarly restrict OBV’s reference.

Regarding the first point, we can note that OBV’s coreference allergy specifically relates to matrix-clause positions. Unlike the behavior with matrix subjects, OBV can corefer with embedded subjects. Consider, for instance, this example from the Atchan New Testament translation:

- (13) Julisi li lephan bɛn atɛ Phólu yí yón, lóhénthé nse nke mónsi salé nke lo wú
 nke jon...
 julisi_i li lep^hã bɛ a-tɛ phólu_j jí pɔ̃, ló-hénthé
 Julius COP person INDF 3.INAN-do Paul thing good DEF-because.of
 n_i-se nke_j mónsi salé [nke_j lo wú nke_j pɔ̃]
 PROX.S.PFV-give OBV permission COMP OBV go see OBV friend.PL
 ‘Julius_i was a person who did good things for Paul_j, so he_i gave him_j per-
 mission for him_j to go see his_j friends...’ (Acts 27:3)

In the final embedded clause, OBV corefers with the local, embedded subject. The constraint on the reference of OBV, therefore, rules out not *all* coreference between OBV and a subject, but rather coreference between OBV and the matrix subject.

Regarding the second point, we can first note that OBV resists being interpreted as coreferential with topics:

- (14) kati_i, mɛ mpɔ mɛ/#nɛ_i ntráljé
 K. 1SG love PROX/OBV clothes
 ‘Katie_i, I love 3SG_i’s clothes.’ (20230624_kou_jea)

Here, we see that OBV resists being construed as referring to Katie. (That is, the disjoint reference effects obtain with a non-subject topicalized nominal.)

One alternative view that we might entertain is that OBV’s orientation is actually purely an anti-topical orientation (cf. Bosch & Umbach 2006, Patel-Grosz & Grosz 2017, a.o., on German demonstrative pronouns). On such a view, we would need to appeal to the idea that apparent anti-subject orientation results when subjects

are topics. However, this view does not extend to all of OBV's disjointness effects. Subjecthood and topicality can be teased apart in sentences with quantified subjects, as shown below:¹²

- (15) δje $brebre_j$ e- $\acute{f}u$ $s\acute{a}l\acute{e}$ $n\acute{e}_{i/*j}$ $p\tilde{o}$
 woman each think COMP OBV pretty
 'Each woman_j thinks 3SG_{i/*j} is pretty.' (20230704_hre)

We observe here that OBV cannot be bound by the quantified subject. In this example, the matrix subject is a quantified expression. Topics are generally assumed to be necessarily referential, with the consequence that quantified nominals are not possible topics (Polinsky & Potsdam 2001). This holds in Atchan of the quantifier *brebre* 'each' discussed here. As shown in (16a), referential nominals in Atchan can be topicalized; here, topicalization is marked by the prosodic break and resumptive pronoun. In contrast, (16b) shows that quantified expressions cannot be topicalized:

- (16) a. $m\acute{o}ja$, $\tilde{a}$ $\eta w\acute{a}$ $gbragbra$
 M. PROX.S run fast
 'Moya, she runs fast.'
 b. * δje $brebre$, $\tilde{a}$ $\eta w\acute{a}$ $gbragbra$
 woman each PROX.S run fast
 Intended: 'Each woman, she runs fast.' (20250326_hre)

Since *brebre*-quantified expressions cannot be topicalized, we can conclude that the disjointness effect in (15) cannot result from a unified anti-topic orientation. Instead, the generalization we have made more cleanly captures this data: high nominals, even when they are not topical, cannot bind OBV.

It is also worth noting that OBV does not track perspective-related notions like logophoric pronouns do. For example, OBV does not track notions like the source of an attitude report, which Sells (1987) shows is relevant for many logophoric elements. Below, Timothée is the source of the attitude report, but OBV exhibits disjointness effects from the structurally-higher subject:

- (17) $kati$ je [$timote$ $m\acute{i}t^h\acute{e}$] $l\acute{e}$ $n\acute{e}$ ge $j\acute{i}$
 K. hear T. mouth COMP OBV win
 'Katie_i heard from Timothée_j that 3SG_{j/*i} won.' (20250326_hre)

In addition, OBV is not sensitive to *de se* orientation, which is relevant for at least some logophoric pronouns (see Anand 2006, Bassi et al. 2023, a.o. for discussion of *de se*-restricted logophors, though cf. Pearson 2015, Akolkar & Hien 2024 for different patterns). Just as OBV cannot refer to the attitude holder *de se*, so too it cannot be used to report a subject attitude holder's *de re* self-belief, as shown below:

¹² This example is discussed in more detail, and with a richer context, in §5.4.2.

- (18) Context: A woman in a tour group in Iceland briefly broke off from the group and changed clothes. She was deemed missing and a search for the lost tourist began. The woman joined the group to search for the lost tourist.¹³

#ŋje lókō ŋu salé nɛ dja
woman DEM think COMP OBV lost

Intended: ‘That woman_i thought that she_i was lost.’ (20230706_hre)

Consultant comment: ‘You can say this sentence in Atchan, but it doesn’t fit the situation.’

Here, we see that a *de re* reading does not permit OBV to corefer with the attitude holder. I conclude that we should not appeal to perspective in explaining OBV’s orientation. Instead, we can more cleanly capture the disjointness effects we see through our generalization that appeals to structural height.

4.2 Reference-tracking effects

In this section, I present additional textual data on PROX and OBV. I show (as briefly mentioned previously) that their alternation is leveraged in the New Testament translation as a form of reference tracking. Examining texts and the reference-tracking effects that emerge in them enables us to corroborate the reference constraint in a new way: we can ask *which referents* in a narrative text are referred to with PROX vs. OBV. What we observe is that OBV is systematically used to refer to less topical entities. Of course, the elicitation data above showed us that the alternation is about more than *only* topicality, but topicality is most salient in this narrative data.

First, we can see reference tracking occur: in the New Testament translation, PROX and OBV are often used in a stable way to refer to contrasting referents in a narrative passage. This is illustrated below:

- (19) Yípo lókōn, nánánmi bɛn bha mɛn pho, nke che ándénga núkhwé, nke hón-
ɛn jika kranmɛn, ló mísé-ndu dɔn mɛn mɛn, nke té yí dri pho...

jípo lókō_i, nánámi bɛ_j ɓa mɛ́_i p^ho, nke_j c^hɛ á_i-ndéŋga
child DEM spirit INDF be.at PROX body OBV cause PROX.S-make.noise
núk^hwé, nke_j hɔ́-ɛ́_i jika krāmɛ́, ló mísé-ndu dɔ́ mɛ́_i mɛ́,
scream OBV take-PROX.O shake hard so saliva hit PROX mouth
nke_j té jí dri p^ho
OBV do thing make.suffer body

¹³ I thank Amy Rose Deal for discussion of this real-world event. Similarly to Pearson 2015:fn.31, consultants read an article about the event before discussion. The article can be found at https://www.francetvinfo.fr/replay-radio/on-ne-pouvait-pas-le-rater/on-ne-pouvait-pas-le-rater-elle-participe-aux-recherches-pour-se-retrouver-elle-meme_3595911.html.

‘That child_i, a spirit_j is in his_i body; he_j makes him_i scream; he_j takes him_i and shakes (him_i) hard, so that saliva foams at his_i mouth; he_j makes (him_i) suffer.’
(Luke 9:39)

This passage opens with an overtly-topicalized individual ‘that child,’ after which a second individual, the spirit that possesses him, is introduced.¹⁴ Throughout the passage, the child is referred to with PROX pronouns, the spirit with OBV. Here, of course, we observe that OBV refers throughout to the non-topicalized individual, in a way that is fully consistent with our distributional and reference constraints. That is, we see evidence that OBV preferentially does not refer to matrix topics in sentences where matrix topics are overtly marked.

We can also observe that the span over which PROX and OBV assignment occurs is smaller than a full text (but, rather, more sentence-level). Specifically, the referents tracked with PROX and OBV can change over the course of a narrative text, in ways that mirror referents’ intuitive topicality. In the following passage, for instance, the forms used to refer to the governor change over the course of the passage:

- (20) *Ŋgbankó-phámán ka adi Sezare khúbhé, wo bho hró se guvɛɛ, ló wo hɔn Phólu se nke. Guvɛɛ kan hró, ló úmi Phólu, nke gotho si nke chan. Lóka úyɛn salé nke chan Sisili gotho ɔn...*
ń-gbankó-phá-má ka a-di sezare kʰúbhé, wo ɓo hró se
 CM-horse-ride-PL time 3.PFV-be Caesarea country 3PL take letter give
guvɛɛ_i, ló wo hɔ pʰólu se nke_i. guvɛɛ_i kã hró, ló
 governor so 3PL take Paul give OBV governor read letter so
ń_i-mi pʰólu_j, nke_j gotho si nke_j cʰã. ló-ka
 PROX.S.PFV-ask Paul OBV country place OBV come.from DEF-time
ń_i-pɛ salé nke_j cʰã sisili gotʰo ɔ...
 PROX.S.PFV-know COMP OBV come.from Cilicia country TOP
 ‘When the cavalry arrived in Caesarea, they gave the letter to the governor_i, and they gave Paul_j to him_i. The governor_i read the letter, so he_i asked Paul_j what country he_j came from. When he_i learned that he_j came from Cicilia...’
 (Acts 23:33-34)

Here, the governor is initially introduced and referred to using OBV. Later in the passage, he is referred to using PROX (and Paul is referred to with OBV).¹⁵ The

¹⁴ It seems reasonable here that the child is the topic of the entire sentence. We could assume that the conjunction here involves conjoining phrases that are smaller than a TopP, or that subsequent clauses contain null topics referring to the child. See §6.2 for more on null continuing topics.

¹⁵ The reader might note that an instance of *guvɛɛ* ‘governor’ occurs between the governor’s reference with OBV and his later reference with PROX. It is compelling to think that this use of an R-expression ‘resets’ the topicality of the governor.

switch in pronouns mirrors a change in the intuitive topicality or discourse centrality of the governor: the governor is originally a less-central individual in the discourse (the previous sentence in this passage focused on the cavalry's travels), but he becomes more central over the course of this passage as he decides what to do with Paul. This supports both the spirit of our reference constraint—OBV is used to refer to entities that are not currently topical—and also a view on which the (im)possible referents of OBV are calculated on a sentence-by-sentence basis.

Additionally, we can observe that the intuitive function of OBV is to mark disjointness from a subject/topic. Two pieces of evidence support this conclusion. First, when a narrative centers on only one referent, only PROX pronouns are used, as shown in the following passage about Jesus:

- (21) $\acute{n}pha$ no ábhóbhó bɛn tho ánná Yankan. Lóka ácí gu, mén míbrɛ $\acute{n}ma$ lóne...
 $\acute{n}_i$ -p^ha no ábóóó bɛ t^ho $\acute{a}_i$ -na jākā. ló-ka
 PROX.S.PFV-climb go mountain INDF top PROX.S-pray God DEF-time
 á-cí gu, mɛ́ $\acute{m}_i$ mí-brɛ $\acute{n}_i$ -ma lóne
 CM-day be.night PROX ?-one PROX.S.PFV-be.at there
 'He_i went to the top of a mountain (and) he_i prayed to God. When it was
 dark, he_i alone was there.'
 (Matthew 14:23)

Here, Jesus is referred to throughout with PROX pronouns. I have found no corresponding passages that center on just one referent and use OBV. That is, OBV is only used when there is contrast between multiple referents. This is consistent with our distributional constraint, which predicts the impossibility of using OBV in passages like this one.

Finally, we note that the translators preferentially use OBV specifically when there are multiple possible third-person referents that can be disambiguated through pronoun choice.¹⁶ This is shown in the near-minimal contrast in (22):

- (22) a. ló ɛdo yípo le mén man ónmɔn wá ló Ejiphithi.
 ló ɛ-do jípɔ_i le mɛ́ $\acute{m}_i$ mã ómɔ wá ló ejip^hit^h_i
 so 2SG.S-receive child with PROX mother with run go Egypt
 'May you take the child_i and his_i mother and flee to Egypt.'
 (Matthew 2:13)
 b. $\acute{n}do$ yípo le nkɛ man ónmɔn no Ejiphithi.

¹⁶ Note, however, that in elicitation examples, OBV can be licensed by speech-act participants:

- (1) mɛ́ mpɔ nɛ wó
 1SG love OBV cat
 'I love 3SG's cat.'
 (20230624_kou_jea)

That is, it is not in general the case that OBV requires a c-commanding *third-person* nominal.

ńj-do jíɔ̃ le nkɛ_i mā ǒmō no ejip^hit^hi
 PROX.S.PFV-receive child with OBV mother with go Egypt
 ‘He_j took the child_i and his_i mother and left for Egypt.’ (Matthew 2:14)

The first sentence, (22a), has only one salient 3SG antecedent for the indicated pronoun, and the translators use PROX to refer to the child. By contrast, (22b) has a different 3SG subject (Joseph, referred to here with PROX). In this sentence, there are two potentially salient referents for the pronoun (either Joseph or the child); here, using OBV to refer to the child resolves this ambiguity, since the subject is a structurally-higher PROX pronoun. These minimally-contrasting sentences clearly show how the alternation between PROX and OBV can be leveraged: because OBV has restricted referential possibilities, its use can limit ambiguity in sentences like (22b).

4.3 Discussion

Through the discussion in this section and the previous one, we have arrived at the conclusion that Atchan OBV is subject to a pair of constraints that restrict its distribution and reference. These constraints are repeated below:

- (23) **Distributional constraint on OBV:** Atchan OBV cannot occur as the highest nominal in the matrix Top > Foc > Subj sequence.
- (24) **Reference constraint on OBV:** Atchan OBV cannot be construed as coreferential with or bound by the highest nominal in the matrix Top > Foc > Subj sequence.

This fundamental pair of constraints feeds into the analysis of obligatorily-bound items. For instance, claiming that an anaphor is subject to Condition A captures both distributional constraints (e.g., anaphors are ruled out in matrix subject position because there is no higher item to bind them) and reference constraints (e.g., obligatory binding/coreference).

This view of disjointness—that disjoint reference effects might come out of obligatory binding—was first suggested by Koopman & Sportiche (1989) for a possibly cognate pair of pronouns in Abé (Kwa, Côte d’Ivoire). In the next section, I develop a semantic account of this phenomenon.

5 An obligatory binding analysis

In this section, I develop a semantic account of obligatory binding of Atchan OBV. The account that I provide is fundamentally semantic, as it relies on a presupposition introduced by OBV and a semantic variable-binding operation. However, the

account does rely on one syntactic component of OBV's orientation, in that the Top > Foc > Subj sequence relevant for OBV's disjointness profile is structurally encoded.

This section proceeds as follows. In §5.1, I show that the disjointness inference associated with OBV projects, supporting an analysis on which it is encoded as presuppositional content. In §5.2, I provide such a presuppositional denotation for OBV, taking inspiration from the presuppositional account of logophoric pronouns developed by Bassi et al. (2023). From there, §5.3 presents the compositional component of the analysis, involving binding of a variable in OBV's presupposition by the highest nominal; §5.4 provides additional examples of bound pronouns and how they interact with OBV's presupposition. From there, §5.5 discusses the distribution of OBV. Finally, §5.6 discusses how long-distance binding at a particular position can be enforced in the semantics.

5.1 Not-at-issue status of the disjointness inference

There is precedent in the literature for formalizing identity and disjointness requirements similar to the one observed in Atchan either as at-issue or not-at-issue content. On the former side, McKenzie (2012) develops a semantic analysis of switch-reference markers that encodes identity or disjointness as at-issue meaning components. The denotations he provides for the same-subject (identity-encoding) and different-subject (disjointness-encoding) morphemes are given below:

- (25) a. Same subject: $\llbracket \text{SR}_{\text{SS}, [m]} \rrbracket = \lambda P. \lambda x. \lambda s. x = g(m) \& P(x)(s)$
 b. Different subject: $\llbracket \text{SR}_{\text{DS}, [m]} \rrbracket = \lambda P. \lambda x. \lambda s. x \neq g(m) \& P(x)(s)$

The contribution of these markers is, roughly, to add an assertion of either identity or non-identity between a lower nominal (the type-*e* argument *x*) and a higher nominal (whose index is *m*).

In a different domain, not-at-issue analyses of pronominal identity requirements have been developed by Sauerland (2013), McKillen (2016), Bruening (2021) for reflexive pronouns and by Bassi et al. (2023) for logophoric pronouns. These authors develop presuppositional accounts of the identity requirements in these domains largely on the basis of ellipsis phenomena. They propose that the identity requirement introduced by reflexive *-self* or logophoric pronouns is presuppositional in nature, since they permit both strict and sloppy readings in ellipsis.

In this section, I argue that the disjointness requirement accompanying Atchan OBV should be modeled as not-at-issue content. The central evidence comes from

projection.¹⁷ As is shown below, the disjointness inference persists in negated sentences:

- (26) *kati_j le pɔ [nɛ_{i/*j} ntráljé]*
 K. NEG love OBV dress
 ‘Katie_j doesn’t like 3SG_{i/*j}’s dress.’ (20250326_hre)

The fact that the disjointness inference persists in this environment provides evidence that it projects.

Another more complex projection pattern occurs in the scope of quantifiers. Here, too, the disjointness inference projects. Consider the following pattern of judgments, a version of (15) with richer context:

- (27) Context: Timothée_k is a salient man, and Lindsay_l is a salient woman.
*ɓje brɛbrɛ_j e-ɓú sálé nɛ_{k/*j/*l} pɔ̃*
 woman each think COMP OBV pretty
 ‘Every woman_j thinks 3SG_{k/*j/*l} is pretty.’ (20230704_hre)

Here, a quantified subject is the highest relevant nominal; binding of OBV by the quantifier is ruled out, as is mere reference of OBV to a woman. That is, each individual picked up by the quantifier must have the property of being disjoint from OBV.

This pattern parallels behavior of gendered pronouns in English, which are standardly analyzed as presuppositional (Cooper 1983, Heim & Kratzer 1998, Heim 2008, Sauerland 2008, Sudo 2012). Projection provides classic evidence in support of a presuppositional analysis of these gender features. This projective behavior can be seen, for example, in the infelicity of sentences like the following:

- (28) [Every man and woman]_j thinks she_{*j} is smart.

Importantly, this sentence cannot mean that *every [man and woman]_j thinks [they_j are smart and they_j are female]*: the gender presupposition projects.

I argue that the parallel between these examples provides evidence that both inferences have a similar presuppositional status. The intuition between the two examples is fully parallel: in (28), binding is ruled out because not every individual meets the presupposition of being female; in (27), not every individual meets the presupposition of being distinct from the referent of OBV. These data will be discussed in more formal detail in §5.4.2.

¹⁷ More investigation of strict and sloppy readings of ellipsis in Atchan is needed before the above-mentioned diagnostic can be applied.

5.2 A two-variable presupposition

We have reached what may look like a contradiction: I have proposed that Atchan OBV is obligatorily bound but is subject to disjoint reference effects. To reconcile these two ideas, I develop a view in which OBV's disjointness inference is encoded presuppositionally via a feature DIFF, which introduces an additional variable. That new variable can be bound separately from the pronominal core, deriving binding with disjointness.

The idea that more complex pronominal forms involve additional variables has had a recent upswing in popularity. This idea emerges in one form, for example, in the work of Schwarz (2009, 2013) on demonstrative pronouns: on his view, certain definite forms introduce index variables and an identity component, while others do not. The account that I develop also has precedent in Bassi et al.'s (2023) account of logophoric pronouns, illustrated below.

Logophoric pronouns are specialized pronouns that occur (roughly) only in attitude complements and, in those environments, necessarily pick out the attitude holder (Hagège 1974, Clements 1974). Perhaps the most famous example of a logophoric pronoun, in Ewe (Gbe, Ghana and Togo), is shown below:

- (29) a. Kofi be yè dzo.
 K. say LOG leave
 'Kofi_i said that he_{i/*j} left.'
- b. Kofi be e dzo.
 K. say 3SG leave
 'Kofi_i said that he_{j/*i} left.'
- (Clements 1974:(142))

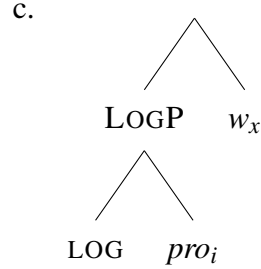
Here, the logophoric pronoun *yè* must refer to the attitude holder in (29a) and contrasts with the behavior of the ordinary pronoun *e* in (29b).¹⁸ Logophoric pronouns have been of continued interest in the semantic literature and are standardly analyzed as involving obligatory binding of the logophoric pronoun (von Stechow 2003, Anand 2006, Pearson 2015, Bassi et al. 2023, a.m.o.).

On Bassi et al.'s account, a logophoric pronoun consists of a pronominal core (for them, a $\langle s, e \rangle$ individual concept) together with a presupposition-introducing feature LOG (30a). The overall denotation that they assign to a logophoric pronoun is presented in (30b) below, with the corresponding structure shown in (30c).

$$(30) \quad a. \llbracket \text{LOG} \rrbracket^g = \lambda f_{\langle s, e \rangle} . \lambda w_x : f(w_x) = x . x$$

¹⁸ Note that the literature diverges as to whether the ordinary pronoun *can* refer back to the attitude holder. Clements and Bassi et al. report that this is impossible, while Pearson (2015) reports that for her consultants, reference of the ordinary pronoun to the attitude holder is dispreferred but possible for at least some speakers.

b. $\llbracket \text{LOGP} \rrbracket^g = \llbracket \text{LOG} \rrbracket^g(\llbracket \text{pro}_i \rrbracket^g) = \lambda w_x : \llbracket \text{pro}_i \rrbracket^g(w_x) = x . x$
 (Bassi et al. 2023:(33-34))



Bassi et al.’s account assumes centered world arguments (w_x), formally an ordered pair including a world and its logophoric center. The contribution of LOG is to wrap the logophor’s pronominal core in an identity presupposition, requiring that the individual picked out by the pronoun in the world of evaluation is indeed the world’s logophoric center x . If that condition is met, the logophor refers to that center.

This presupposition has the effect of adding a new variable, x , into the logophoric pronoun’s denotation. One useful result of this approach is that it makes it possible for the pronominal core to be bound separately from the center variable. This is in fact what happens on Bassi et al.’s account. On their analysis, the pronominal core pro_i often remains free, while the center variable x is always bound at the edge of attitude complements.

In this section, I develop an account of Atchan OBV that similarly involves a two-variable presupposition. The novel contribution of my account in this section is to relate the two variables via non-identity, rather than identity. The intuition behind this account can be traced to Koopman & Sportiche (1989), who propose that a similar (possibly cognate) pronoun in Abé (Kwa, Côte d’Ivoire) exhibits disjointness effects yet is bound. On my account, Atchan PROX is a fully ‘ordinary’ pronoun, while OBV contains both a pronominal core and a presuppositional feature.

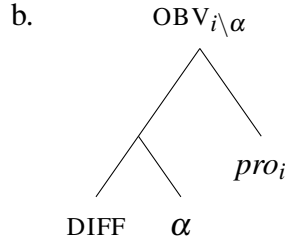
The Atchan pattern does not crucially involve attitude embedding; as such, my account will not involve world variables. Consequently, I adopt a type- e semantics of ordinary pronouns and pronominal cores, as is shown below:

(31) $\llbracket \text{PROX}_i \rrbracket^g = \llbracket \text{pro}_i \rrbracket^g = g(i)$

This formalization follows Heim & Kratzer 1998, though the particular formal choices are not crucial.

The pronoun OBV, on my analysis, consists of a pronominal core with a $\langle e, \langle e, e \rangle \rangle$ feature DIFF. I propose that the contribution of DIFF is introduce a non-identity presupposition between two arguments, as shown below:

(32) a. $\llbracket \text{DIFF} \rrbracket^g = \lambda x. \lambda y : x \neq y . y$



Compositionally, I assume the structure in (32b) above. Notably, the pronominal core here forms the second argument to DIFF (rather than the first argument to LOG on the view sketched above). The view adopted here enables a fuller formal parallel to phi-features, which are standardly assumed to be of type $\langle e, e \rangle$ (Heim 2008, Sauerland 2008, 2013, a.o.): on this view, the sister of the pronominal core is also of type $\langle e, e \rangle$. The negative flavor of DIFF’s nonidentity constraint recalls the work of Hinterwimmer & Bosch (2016, 2017), who propose that German demonstrative pronouns are subject to a semantic constraint which holds that they must be outranked in salience by a different, referentially-distinct nominal in the same sentence.

The first argument to DIFF is an additional variable, written α throughout. The resulting denotation of OBV is given in (33) below:

$$(33) \quad \llbracket \text{OBV}_{i \setminus \alpha} \rrbracket^g = \llbracket \text{DIFF} \rrbracket^g(\alpha)(\llbracket \text{pro}_i \rrbracket^g) = g(i) \text{ defined if } g(\alpha) \neq g(i)$$

The variable α has the effect of restricting OBV’s reference. For ease of reading, I will throughout the paper mark the two elements that factor into OBV’s denotation as subscripts: the first determines OBV’s reference, and the second (always α) occurs only in the disjointness presupposition. Crucially, on this account, OBV’s pronominal core constitutes its asserted content.

On this proposal, OBV comes with a presupposition about the nonidentity of two individuals mapped to by g from i and α . There is evidence that a condition that compares two individuals for nonidentity is correct for Atchan OBV. As is shown below, it is possible for the highest nominal to pick out a plural individual in which the referent of OBV is contained:

$$\begin{array}{ll}
 (34) & \text{wo}_{i+j} \text{ pɔ} \quad \text{nɛ}_i \quad \text{ntráljé} \\
 & \text{3PL} \quad \text{love OBV dress} \\
 & \text{‘They}_{i+j} \text{ like 3SG}_i\text{’s dress.’} \qquad (20240823_hre)
 \end{array}$$

Here, the pronominal core picks up the individual mapped to by i , while α picks up a plural individual containing i . The fact that this is judged acceptable provides evidence that a nonidentity requirement is correct. This would contrast, for instance, with a different view on which total disjointness is required between OBV and the

highest nominal (as is standard in the work on the adjective *different*; cf. Brasoveanu 2011, Hardt & Mikkelsen 2015).

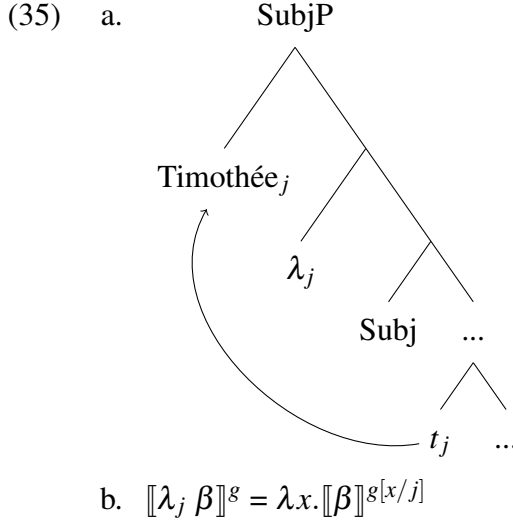
5.3 Capturing the reference constraint: Binding by the highest nominal

Now we will focus on the first argument of DIFF, written α above. This variable α is what restricts the reference of OBV. Our empirical generalization earlier was that OBV cannot corefer or be bound by the highest nominal in the Top > Foc > Subj sequence. Therefore, to capture the generalization, we want α to be bound by that highest nominal (with i free).

How can we capture this binding? Here, again, it is useful to draw a parallel to logophoricity. In the analysis of *de se* logophoric pronouns, the standard approach is to say that the relevant attitude report is *centered* (Heim 2002, von Stechow 2003, Pearson 2015, Bassi et al. 2023, a.m.o., following Lewis 1979). That is, abstraction at the CP periphery targets not only worlds but also the type- e logophoric center of the world. Ultimately, a pronoun referring to counterparts of the attitude holder *de se* saturates that position, yielding the logophoric pronoun's reference to the attitude holder. I propose, for Atchan, that OBV-containing sentences in Atchan similarly involve obligatory abstraction, namely over the variable α by the highest nominal in the Top > Foc > Subj sequence.

Subjects, foci, and topics in Atchan all involve thoroughly ordinary variable binding. Subjects and foci both undergo syntactic movement to their surface positions (Jarvis 2025a,b): these items therefore need at a minimum to bind their traces. In contrast, matrix topics are base-generated in the matrix periphery, but they too can bind lower pronouns. We thus need an account of movement and 'standard' pronominal binding. I adopt an account in the style of Heim & Kratzer 1998, on which movement is semantically interpreted via a coindexed abstraction operator immediately subjacent to the moved element's landing site. This is shown for subject movement in below (I assume that subjects move to Spec,SubjP, though this is not crucial):¹⁹

¹⁹ I write subscripts on all elements that move into these high positions, regardless of whether they are pronouns or R-expressions (contra Heim & Kratzer 1998). For subjects, this is not crucial to the account and is mainly to make clear movement relationships.

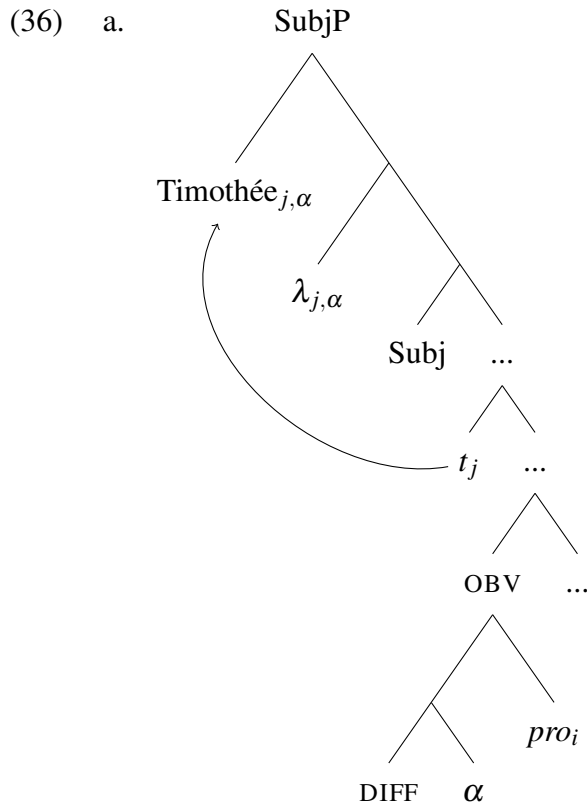


The abstraction operator triggers abstraction over the trace of the moved element and over any coindexed elements through the standard rule in (35b). This semantics for movement and pronominal binding extends in an analogous way to foci (occupying Spec,FocP) and topics (occupying Spec,TopP).

The core of my proposal is that the highest element in the Top > Foc > Subj sequence triggers abstraction not *just* over its trace but *also* over the additional variable α . I propose that the highest nominal in this sequence can bear not one but two index variables, leading to this double abstraction. There are several possibilities for how this double indexing could be. One possibility is that the highest nominal in this sequence comes to gain the feature α over the course of the derivation; this is easiest if we posit some higher (null) head in the syntactic structure. One possibility is that this higher head could agree downward, flagging the highest nominal with an additional feature. Another possibility is that the higher head could syntactically merge with the highest head in the Top > Foc > Asp sequence, endowing this head with an additional feature that it might pass on to the nominal in its specifier. I will not focus on teasing apart these possibilities here.²⁰

This double indexation of the highest nominal is interpreted, I propose, using an operator that abstracts over both indices. This is illustrated below. This structure differs from the previous one in that the subject *Timothée* comes to also bear the feature α , leading to a doubly-indexed operator:

²⁰ The highest nominal in the Top > Foc > Subj sequence could also undergo covert movement to a higher projection whose purpose is to accomplish this double abstraction (though, as a reviewer notes, a movement-based story raises concerns about antilocality and criterial freezing). Note also that it is not crucial whether the trace of the moved element also bears the feature α .



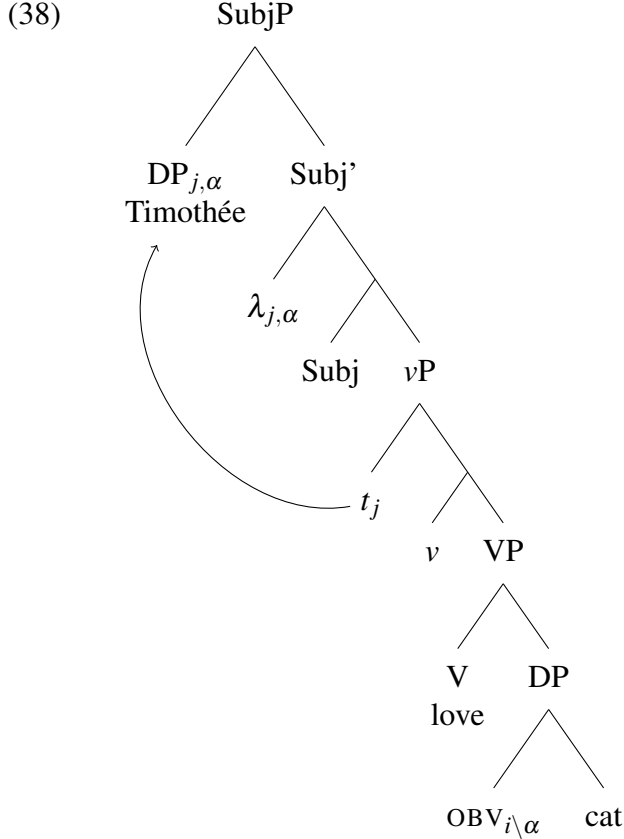
b. $\llbracket \lambda_{j,\alpha} \beta \rrbracket^g = \lambda x. \llbracket \beta \rrbracket^{g[x/j, x/\alpha]}$

The doubly-indexed operator is interpreted as in (36b), causing abstraction over *both* the variable j and the variable α . Note that this doubly-indexed operator is an unselective binder in the sense of Baker (1970), Lewis (1975), Heim (1982), Pesetsky (1987): it binds a specific, multi-membered set of variables, represented here as subscripts on the operator. We will see shortly that it is crucial that only the specific provided set of variables is bound by this operator; some variables (at least the variable i that determines OBV's reference) must not be bound by this operator. This operator is thus unselective in Lewis's sense, in that it binds multiple variables, but not entirely unselective.

This approach gives us a way to capture the anti-prominent orientation of OBV. For a first illustration, let us consider the case of anti-subject orientation of OBV in a simple clause:

- (37) $\text{timote}_j \text{ p}\mathbf{o} \quad \text{n}\varepsilon_{i/*j} \text{ w}\acute{o}$
 T. love OBV cat
 'Timothée _{j} likes 3SG _{$i/*j$} 's cat.'
(20220805_chr_mar)

Here, Timothée is the subject, and no other information-structural marking is present. Therefore, I assume that SubjP is the highest projected phrase in the Top > Foc > Subj sequence, yielding the syntactic structure shown below (showing only relevant projections):



From here, the derivation proceeds as follows. Composition within the VP is standard:

$$(39) \quad \llbracket \text{DP} \rrbracket^g = \iota z [\text{cat}(z) \wedge \text{of}(g(i))(z)] \\ \text{defined if } g(i) \neq g(\alpha)$$

$$(40) \quad \llbracket \text{VP} \rrbracket^g = \lambda x. x \text{ loves } \iota z [\text{cat}(z) \wedge \text{of}(g(i))(z)] \\ \text{defined if } g(i) \neq g(\alpha)$$

The trace of the moved subject saturates the experiencer position:

$$(41) \quad \llbracket \text{vP} \rrbracket^g = 1 \text{ iff } g(j) \text{ loves } \iota z [\text{cat}(z) \wedge \text{of}(g(i))(z)] \\ \text{defined if } g(i) \neq g(\alpha)$$

Then, the doubly-indexed abstraction operator triggers abstraction both over the trace of the moved subject, and over the variable α :

$$(42) \quad \llbracket \text{Subj}' \rrbracket^g = \lambda x. x \text{ loves } \iota z[\text{cat}(z) \wedge \text{of}(g(i))(z)] \\ \text{defined if } g(i) \neq x$$

Finally, the subject in Spec,SubjP saturates this position:

$$(43) \quad \llbracket (38) \rrbracket^g = 1 \text{ iff Tim loves } \iota z[\text{cat}(z) \wedge \text{of}(g(i))(z)] \\ \text{defined if } g(i) \neq \text{Tim}$$

Note that the index i that determines OBV's referent remains free throughout the derivation. Any value of i is therefore licit except values that map to Timothée, which are ruled out by the presuppositional definedness condition. This captures the anti-subject orientation observed in this example. This account extends naturally to sentences involving projected topics and foci, in which case double abstraction takes place at those higher positions. This derives anti-topic and anti-focus orientation of OBV in those cases. In such cases, similarly, OBV's first index i remains free.

Of course, i does not have to be free; as with any other pronoun, the index i can be quantificationally bound. In the next section, I present additional examples involving quantificational binding of i .

5.4 Binding the variable i

Recall from the denotation of OBV, repeated below from (33), that the variable i features in both the assertive and presuppositional meaning components:

$$(44) \quad \llbracket \text{OBV}_{i \setminus \alpha} \rrbracket^g = \llbracket \text{DIFF} \rrbracket^g(\alpha)(\llbracket \text{pro}_i \rrbracket^g) = g(i) \text{ defined if } g(i) \neq g(\alpha)$$

It is worth walking through in detail what happens when the variable i is bound—this occurs when OBV behaves as an ordinary bound pronoun. To treat this formally, we will need entries for quantifiers that can do justice to projective content. An appropriately-projective denotation of *every*, adapted from Heim (2008) and which I will adopt here for Atchan *brɛbrɛ* ‘each’, is given in (45):

$$(45) \quad \lambda f_{\langle e,t \rangle}. \lambda h_{\langle e,t \rangle} : \{x : f(x) \wedge C(x)\} \subseteq \text{dom}(h). \{x : f(x) \wedge C(x)\} \subseteq \{x : h(x)\}$$

The underlined presuppositional component requires that all individuals meeting the restriction of the quantifier satisfy any presuppositions introduced in the quantifier's scope. I have modified this entry slightly from Heim's version to add an explicit contextual domain restriction variable C (cf. von Stechow 1994).

This presuppositional entry is used by Heim to explain the unavailability of binding by the presupposition-introducing pronoun *she* in (28), repeated below:

$$(46) \quad [\text{Every man and woman}]_j \text{ thinks she}_{*j} \text{ is smart.}$$

Gender features in English pronouns are standardly assumed to be presuppositional in nature (Cooper 1983, Heim & Kratzer 1998, Heim 2008, Sauerland 2008, Sudo 2012, a.o.), with *she_j* having a denotation like (47):

$$(47) \quad \llbracket she_j \rrbracket^g = g(j) \text{ defined if } g(j) \text{ is female}$$

Heim shows that the unavailability of binding in (46) follows from (45) and (47). In (46), *she* occurs in the scope of the quantifier *every*. As a result, if *she* here is bound, (45) leads to the presupposition that every individual in the restriction of the quantifier (i.e., every individual in the set of men and women) meet *she*'s presupposition of being female. This is clearly contradictory.

Similar reasoning extends to OBV when its *i* variable is quantificationally bound. To see this, it is worth considering two cases separately.

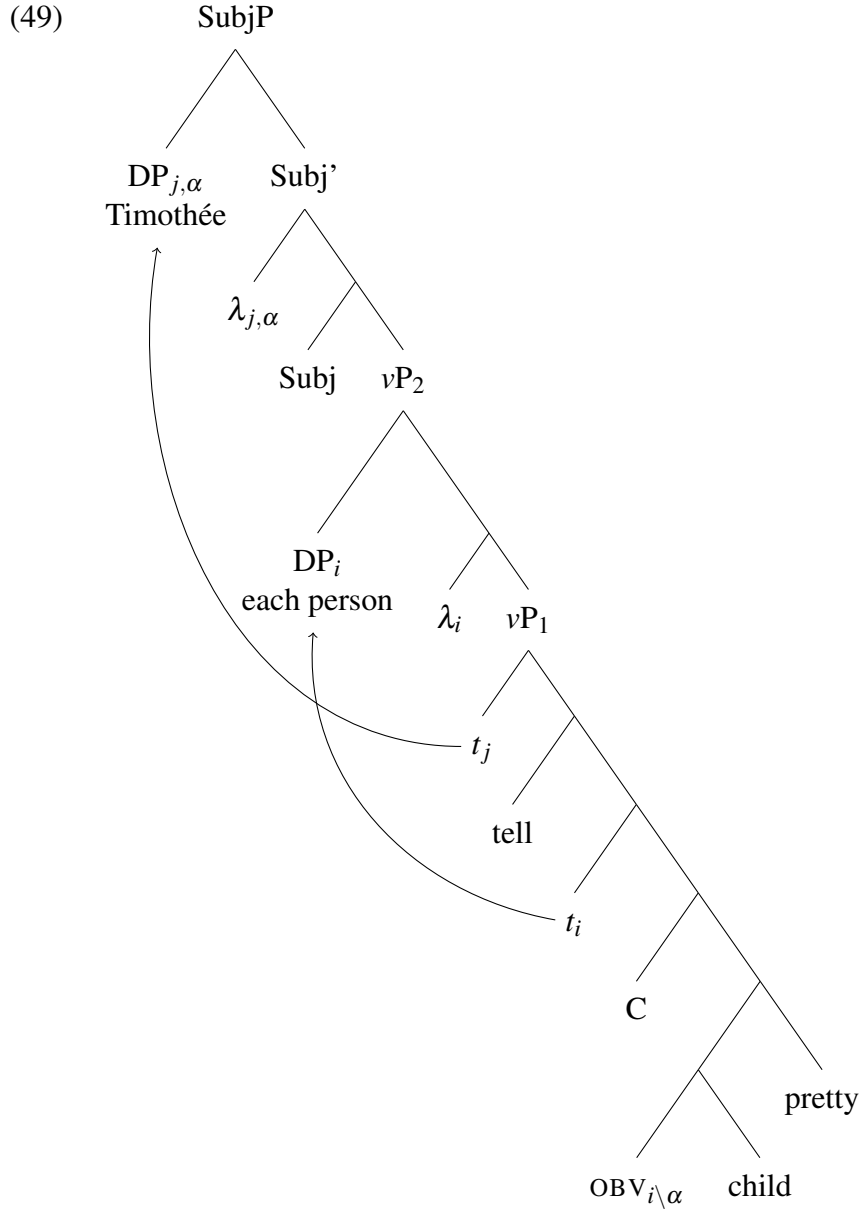
5.4.1 Case I: Quantificational binder does not bind α

The first case to be considered is a case where the quantificational binder of OBV's *i* variable is distinct from the element that binds α . This occurs in (12b), repeated below:

- (48) timote krāmā [lep^hã brebrɛ]_i lê nɛ_i bi pɔ̃
 T. tell person each COMP OBV child pretty
 'Timothée told each person_i that 3SG_i's child is pretty.'
(20230628_hre_chr_mar)

Here, the subject Timothée binds α , while *i* is bound by the lower quantifier.

There are multiple kinds of movement going on in this example: the subject moves to Spec,SubjP, and the quantifier (I assume) undergoes QR. I will treat QR here as adjunction to the lowest type-*t* node in the structure (though this is not crucial) and will indicate the quantifier here as adjoining to ν P. The overall structure that I assume is shown below:



Composition through vP_1 is standard:

$$(50) \quad \llbracket \text{OBV}_{i\backslash\alpha} \text{ child} \rrbracket^g = \iota z[\text{child}(z) \wedge \text{of}(g(i))(z)] \\ \text{defined if } g(i) \neq g(\alpha)$$

$$(51) \quad \llbracket vP_1 \rrbracket^g = 1 \text{ iff } g(j) \text{ told } g(i) \text{ that } \iota z[\text{child}(z) \wedge \text{of}(g(i))(z)] \text{ is pretty} \\ \text{defined if } g(i) \neq g(\alpha)$$

Next, we abstract over the variable i :

- (52) $\llbracket \lambda_i \nu P_1 \rrbracket^g = \lambda x. g(j) \text{ told } x \text{ that } \iota z[\text{child}(z) \wedge \text{of}(x)(z)] \text{ is pretty}$
 defined if $x \neq g(\alpha)$

The quantified DP takes this as its second argument. The quantifier’s presupposition requires that each individual in the domain of the quantifier bear the property of not being identical to $g(\alpha)$. This yields the following:

- (53) $\llbracket \nu P_2 \rrbracket^g = 1 \text{ iff } \forall y[\text{person}_C(y) \rightarrow g(j) \text{ told } y \text{ that } \iota z[\text{child}(z) \wedge \text{of}(y)(z)] \text{ is pretty }]$
 defined if $\forall y[\text{person}_C(y) \rightarrow y \neq g(\alpha)]$

The next relevant operation is binding of j and α together, saturating both positions with Timothée:

- (54) $\llbracket (49) \rrbracket^g = 1 \text{ iff } \forall y[\text{person}_C(y) \rightarrow \text{Tim told } y \text{ that } \iota z[\text{child}(z) \wedge \text{of}(y)(z)] \text{ is pretty }]$
 defined if $\forall y[\text{person}_C(y) \rightarrow y \neq \text{Tim}]$

Let us take stock of what this yields. The asserted content in (54) is what we would expect from any quantificationally-bound pronoun. The definedness condition here requires that each individual being quantified over not be Timothée. This requirement can plausibly be satisfied via domain restriction; the truth of the English translation of (48) does not seem to entail that Timothée has any children.²¹ In sum, what this example shows is that when the nominal that binds OBV’s α is distinct from the nominal that binds i , quantificational binding of OBV is successful, with OBV’s presupposition playing a minimal role.

5.4.2 Case II: Quantificational binder also binds α

However, the situation is quite different if the nominal that binds α is quantificational and also binds i . In this case, quantificational binding is unsuccessful and leads to additional disjoint-reference effects.

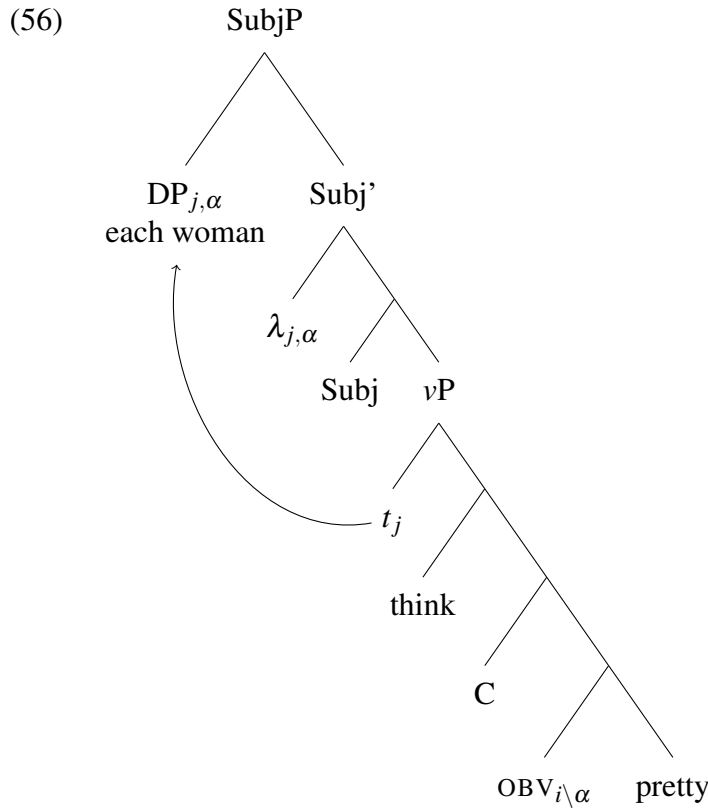
The key data point that we will consider here comes from (27), repeated below:

- (55) Context: Timothée_k is a salient man, and Lindsay_l is a salient woman.
 bje brɛbrɛ_j e-ɓú sálé nɛ_{k/*j/*l} pɓ
 woman each think COMP OBV pretty
 ‘Every woman_j thinks 3SG_{k/*j/*l} is pretty.’ (20230704_hre)

²¹ This is especially satisfiable in the context in which this example was elicited, which involved Timothée talking to a group of mothers_i about their_i children. This facet of the elicitation context is not obvious from the example in (48).

We observe here that binding by the quantifier is impossible, as is mere reference of OBV to a woman being quantified over. This pattern is correctly predicted by this account.

To see this, let us first consider what happens if OBV's i is free. In that case, we have the structure below:²²



Composition through the indicated vP , followed by joint abstraction over j and α , proceeds as in earlier derivations:

$$(57) \quad \llbracket vP \rrbracket^g = 1 \text{ iff } g(j) \text{ thinks } g(i) \text{ is pretty} \\ \text{defined if } g(i) \neq g(\alpha)$$

$$(58) \quad \llbracket \text{Subj}' \rrbracket^g = \lambda y. y \text{ thinks } g(i) \text{ is pretty} \\ \text{defined if } g(i) \neq y$$

The property in (58) then serves as the quantifier's scope argument. As we saw previously, the quantifier's presupposition requires that the definedness condition in (58) hold of each individual in the quantifier's restriction. This yields the following:

²² Here, the result is the same regardless of whether *every woman* undergoes a later step of QR; I omit it for brevity.

- (59) $\llbracket (56) \rrbracket^g = \forall y [\text{woman}_C(y) \rightarrow y \text{ thinks } g(i) \text{ is pretty}]$
 defined if $\{y : \text{woman}(y) \wedge C(y)\} \subseteq \text{dom}(\llbracket \text{Subj}' \rrbracket)$
 i.e. defined if $\forall y [\text{woman}(y) \wedge C(y) \rightarrow g(i) \neq y]$

The presupposition of this full sentence requires that the referent of free OBV not be any contextually salient woman. This derives the restriction of free OBV in (55) to just contextually-salient men, permitting reference to Timothée but not to Lindsay.

We can also consider the case in which OBV's i is bound by the quantifier. As the reader can see, this would trigger abstraction over *both* variables in the definedness condition in (57), giving rise to the presupposition that each woman not be identical to herself (i.e., $\forall y [\text{woman}(y) \wedge C(y) \rightarrow y \neq y]$). This unsatisfiable definedness condition rules out the possibility of quantificationally binding OBV's index i by the same element that binds α .

In summary, we have seen that the two-variable presupposition of OBV permits quantificational binding of OBV by relatively low nominals but not by the highest nominal.

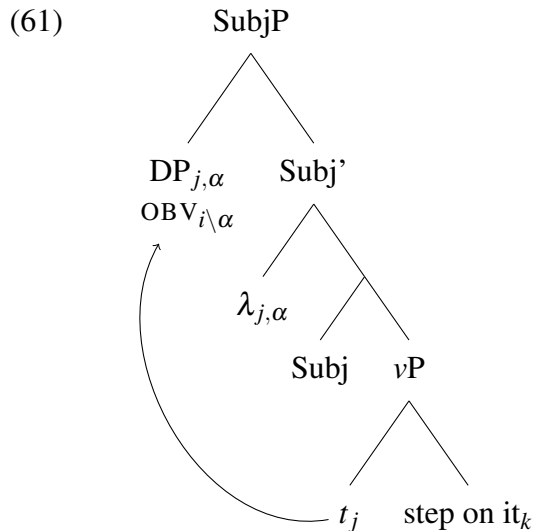
5.5 Capturing the distributional constraint

So far, we have seen that the account developed here can capture OBV's restricted referential possibilities. I show here that the account can also capture its restricted distribution. This follows since only one position can bind α . To fully derive these effects, it is necessary that free α is fatal. With that assumption in place, we can derive the restricted distribution of OBV.

The most obvious way to make free α occur is for OBV itself to *be* the highest nominal. This occurred in (6a), repeated below:

- (60) $\acute{\epsilon}/\#n\acute{\epsilon}$ c^ha ló t^ho
 PROX.S.PFV/OBV step 3INAN top
 '3SG stepped on it.' (20230623_kou_jea)

Because OBV occupies Spec,SubjP, binding of α by below OBV will be vacuous. As a result, free α will survive in the presupposition, as shown below:



- (62)
- a. $\llbracket vP \rrbracket^g = 1$ iff $g(j)$ stepped on $g(k)$
 - b. $\llbracket Subj' \rrbracket^g = \lambda y.y$ stepped on $g(k)$
 - c. $\llbracket (61) \rrbracket^g = g(i)$ stepped on $g(k)$
defined if $g(i) \neq g(\alpha)$

The variable α will also remain free if OBV occurs *inside* the highest nominal. This happens, for instance, if OBV is a subject possessor. Here, OBV is less frequent than PROX, and consultants report the intuition that additional context is needed to make it felicitous:

- (63) mɛ́/?nɛ wɔ̌ pɔ̌
 PROX/OBV cat pretty
 ‘3SG’s cat is pretty.’ (20230625_kou_jea)
 Consultant comment: sentence with OBV is ok ‘if I see someone’

Here, I think we can interpret the comments about particular scenarios permitting OBV as a comment about topicality: OBV can be licensed in this position if something else is topical, but it is certainly not the default. This is correctly predicted by this account, since in this case too binding of α by an operator subjacent to the subject’s surface position will be vacuous, and free α will survive to the end of the derivation.

5.6 Discussion: Whence *obligatory* binding?

The account developed here clearly depends on α being bound at the right spot: to capture our empirical generalizations, it is important both that α not be free, and that nothing other than the highest nominal bind α .

The requirement that most elements in the sentence *not* bind α recalls Heim’s (1983) novelty condition on indefinites, formalized as a requirement that the index associated with the indefinite not be in the domain of the context-provided assignment function g . Of course, however, we need to say more, since it is necessary that α eventually be bound. Another idea that this constraint recalls is the view that Koopman & Sportiche (1989) sketch for a disjointness-marking pronoun in Abé. They posit that this particular pronoun has a special status—namely, that of a ‘logical variable’, which mandates that it be bound. Making this claim encodes half of the generalization: this element cannot be free because it must be bound. However, it is not necessarily clear what the theoretical status of logical variables is, and it is also not obvious why only certain elements should be able to bind this logical variable. (Koopman and Sportiche explicitly leave the question of determining what binds the relevant variable in Abé to future work.)

I would like to adopt the intuition of the ‘logical variable’ view but make the reasoning more precise. Specifically, I propose that we should adopt a more nuanced view of the assignment function g . We might sometimes assume that g can apply to any element of the correct type, but this need not be the case. I follow Kratzer (1991) in positing that variables of a given type can be split into two varieties: *ordinary* variables, which are in the domain of the matrix assignment function g , and *distinguished* variables, which are not. Kratzer develops this idea in an analysis of focus features, but I suggest that we should appeal to a similar notion here.

I propose that OBV’s reference-restricting variable α is a distinguished variable, hence guaranteed not to be in the domain of the context-provided matrix assignment function g .²³ This captures half of our generalization: if α is not in the domain of g , then g is unable to map α to an individual, and hence free α is not well-defined. Since the standard way to accomplish pronominal binding is to replace one index in the assignment function with another, this will also make it difficult for other elements to bind α .

How, then, can α be bound? What I would like to suggest is that *only* the abstraction introduced by the highest nominal is able to relate between ordinary and distinguished variables to bind α . To be fully concrete, this might require a somewhat specific contribution of the doubly-indexed abstraction operator. Adopting a notational convention from Wold (1996), we could define this operator as below:

$$(64) \quad \llbracket \lambda_{j,\alpha} \beta \rrbracket^g = \lambda x. \llbracket \beta \rrbracket^{g[x/j] \cup \{ \langle \alpha, x \rangle \}}$$

23 Note that it is possible for there to be multiple type- e distinguished variables; DIFF could in principle introduce as its first argument any of these distinguished variables. If the distinguished variable introduced by OBV is ultimately bound, the derivation will succeed; in contrast, if the distinguished variable introduced by DIFF is not ultimately bound, the derivation will crash. In this chapter, I invoke exactly one distinguished variable, α , for convenience.

This operator thus maps ordinary variable j to x by partially overwriting the assignment function in the usual way, and also adds a wholly new entry to the assignment function to map α to x . The rest of the sentence is then evaluated not with respect to contextually-given g —on which α was undefined—but with respect to this newly-expanded assignment function. On this view, then, the obligatory-binding profile of OBV results on one hand from the distinguished status of α and on the other hand from the limited ways in which α can be rendered interpretable.

6 Predictions and consequences

This section focuses on exploring some predictions and consequences of the analysis. I discuss the relevance of c-command in §6.1, the possibility of null peripheral structure in §6.2, and *Maximize Presupposition!* in §6.3.

6.1 Predictions: Teasing apart high positions and c-command

It is worth explicitly noting that, on this analysis, binding of α requires c-command. This is because the only nominal that can bind α occupies (via movement, for subjects and foci, or base-generation, for topics) the specifier of a relatively high functional projection.

This prediction that disjointness inferences and c-command correlate is borne out when we look at structurally-complex matrix subjects. In these sentences, OBV is forced to be disjoint from the overall subject but can corefer with other nominals contained within the subject. For instance, lower OBV in (65a) can refer to Katie:

- (65) a. [kati_i mi bje]_j e-bú sálé ne_{i/*j} pĩ
 K. child girl prog-think COMP OBV pretty
 ‘Katie_i’s daughter_j thinks 3SG_{i/*j} is pretty.’ (20230705_hre_chr_mar)
- b. [se k^hé kati_i wu]_j ã_j mú kákólé ne_{i/*j} pĩ
 man COMP K. see PROX.S think COMP OBV pretty
 ‘The man that Katie_j saw thinks that 3SG_j is pretty.’ (20230704_yap)

This is predicted on our account: it is (rigidly) the highest nominal, in (65a) Katie’s daughter, from which lower OBV must be distinct. In this way, binding of α differs from ordinary quantificational binding, which has been argued not to require c-command (Bach & Partee 1980, Barker 2012, a.o.):

- (66) [[Every boy]_i’s mother] loves him_i.

Here, a quantifier inside the subject can bind a pronoun that the quantified nominal does not c-command. This binding is analogous to the coreference observed in (65). This data underscores that binding of α is fundamentally different than binding of ordinary indices.

6.2 Null peripheral elements

In the presentation of the analysis to this point, I have focused on examples where it is straightforward to overtly identify the highest projected nominal in the Top > Foc > Subj sequence. In fact, however, there is evidence that null topics can bind α . The evidence comes from reference-tracking situations, where, as it turns out, OBV can sometimes occur as a subject with no overt higher structure. An example from a story told during elicitation is provided below:²⁴

- (67) Context: Yesterday, Katie_j did something bad. PROX_j hit Timothée_i.
 ént^hé nɛ_i c^hrwe_j
 now OBV hit.PROX.O
 ‘Then, 3SG_i hit 3SG_j.’ (20230823_hre)

Here, OBV is licensed in matrix-subject position even though no higher nominal is visible.

To account for the felicity of examples like (67), I assume that a null topic pronoun referring to Katie can be projected in narrative sentences with continuing topics. That syntactically-projected null topic is thus the highest nominal in the sequence, so the null Top head binds α . Assuming that Katie is the topic throughout this narrative sequence, this sentence is therefore correctly predicted to be felicitous, since Timothée, the referent of OBV, is not coreferential with the topic.²⁵

This observation raises the question of how these null topic pronouns might distribute. My explorations thus far suggest that null topics are likely only possible as continuing topics (which Aissen (2017) notes are often reduced or null cross-linguistically). Additionally, I suspect that there might be a dispreference for vacuous topicalization, i.e., a dispreference for null topics with no resumptive correlates. A fuller exploration of when null topics can occur must be left to future work.

6.3 On Maximize Presupposition effects

I now turn to PROX, OBV’s counterpart. Our discussion of PROX has made it clear that it can, but need not, corefer with higher items. This optional coreference was shown in (1a), repeated below:

- (68) timote_j pɔ mɛ^h_{i/j} wɔ
 T. love PROX cat

²⁴ The verb in (67) is [c^hrwe] ‘hit.PROX.O’, underlyingly /c^hrwa-ɛ/ ‘hit-PROX.O’.

²⁵ The positing of null topics in Atchan might be surprising from an areal perspective. Niger-Congo languages do not generally permit pro-drop (Bisang 2014). However, cross-linguistic evidence suggests that null topics can occur in non-pro-drop languages. For example, German ‘topic drop’ is standardly analyzed as involving null topic pronouns in German, an otherwise non-pro-drop language (Fries 1988, Trutkowski 2016, a.o.).

‘Timothée_j likes 3SG_{i/j} cat.’

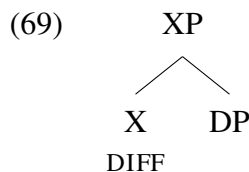
(20220805_chr_mar)

This optionality is captured by the semantics I have proposed, since any ordinary pronoun can optionally be coindexed with other elements.

However, an important question remains: why does *Maximize Presupposition!* (henceforth *MP*, Heim 1991) not always force coreference between PROX and the structurally highest item? On this account, OBV has an extra presuppositional component that PROX does not. Since *MP* generally requires that presupposition-carrying items be used instead of other options whenever the relevant presupposition is satisfied, the corresponding logic would suggest that presupposition-bearing OBV must be used whenever its disjointness presupposition is satisfied. The effect would be that OBV should be the only 3SG pronoun option when that pronoun is disjoint from the highest nominal. That is, lower PROX should obligatorily corefer with the highest nominal. This is not actually the case, as seen in (68). I think there are at least two possible explanations of this fact, which I explore in turn.²⁶

One option would be to claim that PROX and OBV are not *MP* competitors due to structural differences (Katzir 2007, Fox & Katzir 2011). Specifically, we could posit that PROX and OBV have syntactic differences, due to OBV’s additional DIFF feature.

Gender presuppositions are often assumed to be introduced in one of two locations in a pronoun: either they adjoin to the pronominal DP (Heim & Kratzer 1998, Heim 2008), or they are hosted in a Φ P that c-commands DP (Sauerland 2008). In parallel with the latter approach, we could assume that DIFF is hosted in a higher head X above DP:



We could then argue that PROX and OBV do not compete precisely because OBV contains more syntactic structure than PROX.

Another possibility would be to assume some Atchan-specific reason why PROX and OBV do not compete. The most salient possible explanation relates to Dido’s (2018a) observation that OBV is used less by younger speakers: from this observation, we might assume that a language change is currently in progress in Atchan, in which OBV is being lost.²⁷ Speakers who are losing OBV might reanalyze [n(k)ε]

²⁶ There is also the possibility that *MP* might not be equally strong across all languages and phenomena (cf. discussion in Hohaus & Bowler 2022). Exploring this would require more investigation of other domains in Atchan like definiteness and tense where we would expect *MP* effects.

²⁷ As mentioned in Footnote 3, the apparent plural counterpart to OBV, [ɲkɔ̃], has already been lost by many speakers. Of my six consultants, five do not recognize the form [ɲkɔ̃]; the sixth recognizes it

as one object form of the 3SG pronoun; coincidentally, this aligns with early case-based descriptions of Atchan pronouns, like the description of Bôle-Richard (1983).

One empirical advantage to positing this change-in-progress is that it might help to explain why speakers often are more willing to permit coreference between high nominals and $[n(k)\varepsilon]$ if the lower $[n(k)\varepsilon]$ occurs in direct-object position. This was shown, incidentally, in the contrast between (6c) and (14), which are repeated below with different emphasis:

- (70) a. #kati_i, mē mpo ne_i ntráljé
K. 1SG love OBV clothes
Intended: ‘Katie_i, I love 3SG_i’s clothes.’ (20230624_kou_jea)
- b. elize/#ne_i, mē nŏ mē ndo mã ne_i mŏ
L./OBV 1SG FOC 1SG help OBV PART
‘Lindsay/#3SG_i, it’s me who helped 3SG_i.’ (20230628_hre_chr_mar)

This pair of data points reveals multiple facts. First, speakers reject OBV as an overt matrix-clause topic, as shown in (70b). Similarly, they reject OBV as a resumptive element for non-DO items, like the possessor in (70a), but sometimes permit DO resumption of topics by [n(k)ε] in (70b). In my data, this exceptional [n(k)ε]-resumption happens only when the topic is resumed by a direct-object pronoun. This [n(k)ε]-resumption is not explained on the account developed to this point; appealing to a change-in-progress could explain why [n(k)ε]-resumption is sometimes exceptionally permitted specifically in direct-object position.

7 Syntactic licensing accounts

The account that I have developed in this paper is fundamentally semantic and permits an unbounded, nonlocal relationship between OBV and the higher nominal from which it must be distinct. In this section, I contrast this account with a set of more syntactic proposals for capturing disjointness effects. I discuss the prospects of appealing to Binding Conditions in §7.1 and to Agree or covert movement in §7.2, showing that we do not find evidence to support these approaches.

7.1 Binding Conditions

In the syntactic literature, the most familiar way to capture disjointness effects comes from classic Binding Conditions (Chomsky 1981). Namely, both Condition

only as a form used by older generations. Thus, it seems that the pronominal system of Atchan is generally in flux at present.

B and Condition C are ‘about’ disjointness: as a result, we might want to posit that Atchan OBV is subject to one of these conditions. In this section, I show that neither of these conditions, as they are traditionally defined, can capture the behavior of OBV.

First, I consider the possibility that OBV might be subject to Condition C, i.e., that it must be globally free. This idea has precedence, for instance, in [Déchaine & Wiltschko’s \(2002\)](#) claim that certain proforms behave like R-expressions in being subject to Condition C. We have already seen evidence against this, since OBV can corefer with (somewhat) higher c-commanding elements. Textual data on embedded subjects is shown in (71a), and elicitation data on the addressee of *tell* in (71b) (repeated from (13) and (12a) respectively):

- (71) a. Julisi li lephan bən até Phólu yí yón, lóhénthé nse nke mónsi salé nke lo
wú nke jón...
julisi_i li lep^hã bẽ a-té phólu_j jí jǒ, ló-hénthé
Julius COP person INDF 3.INAN-do Paul thing good DEF-because.of
ń_i-se nke_j mónsi salé [nke_j lo wú nke_j jǒ
PROX.S.PFV-give OBV permission COMP OBV go see OBV friend.PL
]

‘Julius_i was a person who did good things for Paul_j, so he_i gave him_j permission for him_j to go see his_j friends...’ (Acts 27:3)

- b. timote krāmā lízi_i lê nε_i fi jǒ
T. tell L. COMP OBV child pretty
‘Timothée told Lindsay_i that 3SG_i’s child is pretty. (20230629_yap)

In each of these examples, OBV corefers with an element that c-commands it, providing evidence against a Condition C-based account.

Next, I turn to Condition B. A Condition B-based generalization would amount to claiming that the lower positions where OBV occurs are in the same binding domain as the relevant higher nominal.

To investigate this, we must establish what constitutes a binding domain for Condition B in Atchan. We are in luck: Atchan PROX obeys Condition B. As shown below, object PROX cannot corefer with the local subject:

- (72) káti pw-ε
K. love-PROX.O
‘Katie_i loves 3SG_{*i}.’ (20250326_hre)

As such, we can identify Condition B binding domains by investigating the coferential properties of PROX: if PROX, occupying a particular position, is able to corefer with the highest nominal, we can conclude that that particular position is in a distinct binding domain from that of the highest nominal.

This diagnostic shows us that, for example, embedded-clause subjects do not occupy the same binding domain as the matrix subject, since a PROX embedded subject can corefer with a matrix subject. This data was shown in (11a), repeated below with binding domains schematized:

- (73) (timote bu)_{D1} (sákolé ã má má)_{D2}
 T. think COMP PROX.S FUT come
 ‘Timothée_j thinks that 3SG_{i/j} will come.’ (20220805_yap)

Having concluded that the embedded subject and matrix subject occupy different binding domains, we can return to the behavior of OBV in the embedded subject position. The crucial data comes from (11b), repeated below with schematized binding domains:

- (74) (timote bu)_{D1} (sákolé nɛ ɓa ɓá)_{D2}
 T. think COMP OBV FUT come
 ‘Timothée_j thinks that 3SG_{i/*j} will come.’ (20220805_yap)

In (74), the matrix subject *Timothée* occupies a different binding domain than the lower OBV pronoun. Therefore, the disjoint reference effect we observe in this example cannot straightforwardly result from Condition B—here, at least on the surface, there is no c-commanding element in OBV’s binding domain that could give rise to this disjointness effect.

Of course, there is always a possibility that the attitude complement may contain null structure, especially including a null representation of the attitude holder (as in Charnavel 2019). This kind of approach holds that the lower binding domain D2 is more complex, containing (say) a null PRO referring to Timothée. If this is right, then the observed disjointness effect could follow as a Condition B effect.

While that explanation could apply to (74), it cannot fully explain the behavior of OBV. This is because not all positions where OBV’s disjointness effect obtain are ones where it is natural to posit null structure. In the next section, for example, we will see that OBV exhibits disjoint reference effects in temporal clauses. To my knowledge, there is no evidence of this kind of null perspective-related structure in such clauses.

In closing, I note that, in this way, Atchan OBV differs from the phenomenon also sometimes termed ‘obviation’ in Romance and other European languages (Piccolo 1985, Kempchinsky 1986, Farkas 1992, Quer 2017, a.o.). In that literature, ‘obviation’ specifically refers to disjoint reference effects that involve embedded-clause *subjects*. Since it only involves subjects, Condition B has been an attractive analytical avenue (see Stegovec 2019 for a recent analysis that extends this line of analysis to some additional matrix-clause phenomena). Given this difference in empirical behavior, it is not clear that these phenomena should be descriptively or analytically unified.

7.2 Agree or covert movement

In this section, I turn to two kinds of accounts that would posit some syntactic dependency between OBV and (a position close to) the highest nominal: an Agree relation between OBV and the highest nominal, or covert movement of OBV to a position near the highest nominal.

The first account that I consider in this section hold that referential dependencies between two nominals are mediated by Agree. Recently, this approach has especially gained prominence in the literature on (canonical) switch reference. In switch-reference phenomena (so named by [Jacobsen 1967](#)), information about (non)coreference of clausal subjects is tracked by specialized markers that occur in the clausal spine. In a typical switch-reference system, one form of the marker (termed ‘same-subject’) surfaces when the subjects of the two clauses corefer, while a different form (termed ‘different-subject’) surfaces when the two subjects do not corefer.

The last several years have seen a growth in accounts of switch reference that invoke the operation Agree ([Baker & Camargo Souza 2020](#), [Arregi & Hanink 2021](#), [Clem 2023](#)²⁸). On these accounts, the coreference comparison between the two relevant nominals is accomplished by means of Agree. Namely, the switch-reference-hosting head (or heads, on the proposal of [Baker & Camargo Souza 2020](#)) establishes an Agree relation with each of the two subjects. On each of these accounts, those Agree relations must be established for switch-reference morphology to successfully surface (though the details on how the Agree relationships relate to surface switch-reference morphology varies across accounts, and is not crucial for our purposes).

We might imagine developing an account that similarly depends on Agree for Atchan. In the domain of pronoun-licensing phenomena, one salient parallel comes from [Pearson \(2015\)](#), who proposes that a logophoric pronoun must be Agreed with and thus licensed by a higher operator. Adapting the account to Atchan, we might claim that some head X, perhaps located near the highest nominal, probes downward and Agrees with OBV.²⁹ For example, for the example in (1b), repeated as (75a) below, we might claim that an Agree relation obtains between X and lower OBV as schematized in (75b):

28 I introduced [McKenzie’s \(2012\)](#) more semantic approach to switch reference in §5.1. As [Arregi & Hanink \(2021\)](#) note, his account ultimately makes even more stringent predictions about locality than the accounts discussed at present.

29 There are a few choice points in this sketch that are not crucial for our purposes. On one hand, for the argument I make here, it does not matter if X also Agrees with the highest nominal. It also does not matter if X is located near the highest nominal (and probes downward) or if X is located near OBV (and probes upwards).

- (75) a. timote_j pɔ̃ nɛ_{i/*j} wɔ̃
 T. love OBV cat
 ‘Timothée_j likes 3SG_{i/*j}’s cat.’ (20220805_chr_mar)
 b. [Timothée X love OBV cat]
 └───┬───┘

On this view, the Agree relation shown here must be established for OBV to be licensed or for the disjointness effects to obtain.

The second account under consideration would capture the disjointness effects observed with OBV by positing that it undergoes additional covert movement, bringing it into the binding domain of the subject. This is the kind of analysis that [Hestvik \(1992\)](#) proposes for an anti-subject-oriented pronoun in Norwegian. On such a view, one would claim that the behavior of OBV does, in fact, result from Condition B, but that the calculation of Condition B effects takes place only after OBV undergoes covert movement into the highest nominal’s binding domain.

Both the Agree-based approach and the covert movement approach predict that we should see locality effects between OBV and the highest nominal. In particular, they both predict that CP-containing strong islands should disrupt the obligatory-disjointness relationship between OBV and the highest nominal, because CP and island boundaries should block the establishment of the relevant dependencies between the two elements. On the Agree-based front, this locality prediction is discussed explicitly in the switch-reference literature ([Baker & Camargo Souza 2020](#), [Arregi & Hanink 2021](#)). These authors show that the nominal expressions being compared in the switch-reference systems they analyze cannot be separated from the probing head by an additional phase head, where at least C and perhaps *v* are phase heads. On a covert movement story, the prediction about locality results because strong islands should block covert movement of OBV.

This prediction about locality is not borne out. To see this, we will look at temporal clauses. As seen in (76b), these clauses in Atchan are strong islands:

- (76) a. mẽ mp^{hi} dja [lɔ̃ ka k^{hɛ́} ɛlize srɛ]
 1 SG laugh DEF time COMP L. stumble
 ‘I laughed when Lindsay stumbled.’
 b. * thábẽ_i nɔ̃ ɛ p^{hi} dja [lɔ̃ ka k^{hɛ́} t_i a srɛ]
 who FOC 2SG.S laugh DEF time COMP ASP stumble
 Intended: ‘Who_i is such that you laughed when they_i stumbled?’ (20250326_hre)

Note that, in addition to being strong islands, these temporal clauses also contain an overt complementizer and are syntactically large.

Contrary to what these accounts predict, (77) below shows that OBV can occur inside this island and exhibits its usual disjointness effects in this environment:

- (77) kati_i p^{hi} dja [ló ka k^{hɛ́} nɛ_{*i} srɛ]
 K. laugh DEF time COMP OBV stumble
 ‘Katie_i laughed when 3SG_{*i} stumbled.’
 (20230703_kou_jea, 20230704_yap)

This shows that the locality prediction made by the Agree-based and covert movement accounts does not hold up.

I thus conclude that we cannot find evidence to support these syntactic approaches to referential restrictions, rendering them less attractive. In contrast, the semantic approach that I have developed in this paper can capture the data seen in (77), as it permits a truly non-local relationship between OBV and the higher nominal.

8 Situating Atchan obviation in a landscape of reference-tracking phenomena

In this section, I briefly discuss comparisons between the Atchan pronominal alternation and other reference-tracking phenomena across languages. We have already seen comparison to logophoric pronouns in §5, Romance obviation in §7.1, and switch-reference phenomena in §7.2. In this section, I briefly discuss Algonquian obviation in §8.1, demonstrative pronouns in §8.2, and strong and weak pronouns in §8.3.

8.1 Algonquian obviation

I have chosen to call the specialized pronoun in Atchan an ‘obviative’ form, drawing a terminological parallel to obviation systems in Algonquian languages.³⁰ In Algonquian-style obviation systems, distinct morphology tracks different referents in a sentence or discourse on the basis of prominence (Bloomfield 1946, Rhodes 1990, Aissen 1997, Thomason 2003, a.m.o.). In Algonquian, this contrasting morphology occurs on both pronouns and lexical nominals.

Will Oxford (p.c.) notes that, in at least some Algonquian languages, this alternation can give rise to disambiguation effects that parallel the ones observed in Atchan. For example, in the following Southwestern Ojibwe examples, we see that two proximates can (and usually do) corefer, while a proximate and an obviative cannot corefer:³¹

30 I thank Will Oxford and Peter Grishin for discussion of Algonquian obviation and the material in this section.

31 For readability, I write a null possessor in these examples and add subscripts to index obviation status.

- (78) a. $Rose_{PROX}$ o-wāpamān [pro_{PROX} o-māmāyan_{OBV}]
 Rose 3-see.DIR.OBVP 3-mother.OBV
 ‘Rose_i sees her_{i,(j)} mother.’
 b. $Rose_{PROX}$ o-wāpamān [pro_{OBV} o-māmāyini_{OBV}]
 Rose 3-see.DIR.OBVP 3-mother.further.OBV
 ‘Rose_i sees her_{j/*i} mother.’ (Schwartz & Dunnigan 1986:(101)-(102))

In both of these examples, the subject *Rose* is proximate-marked, and the object possessum *mother* is obviative-marked. The crucial alternation concerns whether the null possessor is proximate- or obviative-marked. In (78a), the null possessor is proximate-marked (this is indicated by the presence of ‘normal’ obviative marking on the possessum); here, just as in Atchan, coreference between the subject and the possessor is common but not obligatory. In contrast, in (78b), the null possessor is obviative-marked, as is indicated by ‘further obviative’ marking on the possessum (Wolfart 1978, Oxford 2017a). This obviative-marked possessor cannot corefer with the proximate-marked subject.

On the analytical side, the mechanism of obligatory binding has also been invoked in the analysis of Algonquian obviation by Branigan & Mackenzie (1999) and Oxford (2017b). Notably, Oxford (2017b) proposes that the semantics of obviation involves a kind of unselective binding. On his account, a structurally-high proximate-marked element attempts to bind all nominal expressions in its c-command domain. Crucially, obviative-marked elements contain additional syntactic structure that renders them unbindable (for Oxford, this extra structure is KP, inspired by Richards 2010). As a result, all proximate-marked elements are bound (and therefore ultimately corefer with the high element), while obviative-marked elements ultimately come out as disjoint. The long-distance binding of this analysis has clear parallels to the analysis that I have proposed for Atchan. I leave for future work an exploration of the degree to which an Atchan-style analysis could be extended to Algonquian obviation.

8.2 Demonstrative pronouns

Many researchers have noted that in languages like German, pronouns that are build out of demonstratives exhibit a range of special effects, including coreferential restrictions (Wiltschko 1998, Bosch & Umbach 2006, Schwarz 2009, 2013, Hinterwimmer & Bosch 2016, Patel-Grosz & Grosz 2017, Sichel & Wiltschko 2021, a.o.). Here, I highlight three empirical differences between Atchan OBV and German-style demonstrative pronouns.

First, a brief note: the form of Atchan OBV is not related to demonstratives. Atchan has a demonstrative, [lókō], with both adnominal and pronominal uses:

- (79) mē mpo [a-wó lókō] brē [lókō]
 1 SG like CM-cat DEM exceed DEM
 ‘I like this cat better than that (one).’ (20210905_kou_jea)

In this example, *a-wó lókō* ‘this cat’ shows its adnominal use, and later *lókō* shows its pronominal use. Canonical demonstrative pronouns are, unsurprisingly, morphologically identical to (or at least built out of) demonstratives, but OBV and [lókō] are fully morphologically distinct.

Second, an empirical difference: German demonstrative pronouns presuppose that the pronoun’s referent has a linguistic antecedent (Schwarz 2009, Patel-Grosz & Grosz 2017). This rules out, for example, uses of German demonstrative pronouns with implicit antecedents (like the child implicit in the term *pregnant*):

- (80) Wenn ich schwanger werde, werde ich {es/#das} auf jeden Fall behalten.
 if I pregnant become will I it/DEM on every case keep
 ‘If I get pregnant, I will definitely keep it (= the baby).’
 (Patel-Grosz & Grosz 2017:34a)

By contrast, Atchan OBV can refer to implicit antecedents. The relevant data is shown below:

- (81) a. k^hé fje t^hi lábe, ã mú [mé fi] hémpe
 if woman ripen belly ME.NOM think PROX child thought
 ‘If a woman is pregnant, she thinks about her child.’
 b. k^hé fje t^hi lábe, ã mú [nɛ] hémpe
 if woman ripen belly ME.NOM think OBV thought
 hre: ‘If a woman is pregnant, she thinks about ‘the other person’—
 maybe it’s her child? maybe her husband?’ (20230802_hre)

Here, (81a) is a baseline sentence with no implicit reference; PROX in this sentence is a donkey anaphor referring back to the woman. Our goal is to see whether an obviative pronoun in the same position as [PROX child] can refer to the implicitly-mentioned child. This data is shown in (81b). Unsurprisingly, consultants note that these sentences are highly referent-ambiguous, as the consultant’s comment suggests. The crucial judgment here, however, is that the unborn child is a possible referent of OBV in (81b): it does not seem that explicit prior linguistic reference to the referent of OBV is required.³²

Finally, German demonstrative pronouns are proposed to take on an anti-topic orientation (Bosch & Umbach 2006). However, as we have seen, Atchan OBV does not take on a unified anti-topic orientation. Instead, non-topical subjects and focused items resist OBV coreference just like topicalized items do.

32 Of course, in this data it is difficult to know how much consultants are accommodating these sentences by imagining additional context.

8.3 Strong and weak pronouns

It might be intuitive to think of the Atchan alternation as involving a contrast in pronominal strength: perhaps OBV is ‘stronger’ and therefore resists coreference. This idea that strength relates to some of these patterns is pervasive, with echoes in the work on demonstrative pronouns discussed above.

I briefly note here, however, that the Atchan alternation cannot be assimilated to the strong-weak pronominal distinction in [Cardinaletti & Starke \(1999\)](#). As it turns out, both PROX and OBV meet the syntactic criteria that Cardinaletti and Starke associate with strong pronouns. We have seen previously that both can occur in at least certain peripheral positions: PROX in (6b) and OBV in (7b), among other places. Both can also be coordinated:

- (82) mē ŋwu mḗ léka ne
 1SG see PROX or OBV
 ‘I saw either 3sg_i or 3sg_j.’ (20220811_yap)

This data shows that we cannot, for instance, treat PROX as Cardinaletti and Starke’s weak pronoun and OBV as their strong pronoun. Instead, insofar as strength might be a useful way to conceptualize this pronominal contrast, the Atchan data call for a more nuanced view of pronominal strength and the factors that impact it.

8.4 Discussion

This discussion has focused on contrasting the Atchan pronominal system and the reference-tracking effects that arise in it to intuitively-similar reference-tracking phenomena in other languages. In general, we have seen that despite obvious empirical similarities, analyses of other reference-tracking phenomena cannot be extended to Atchan in full. Instead, at least for now, I suggest that we should think of these all as a constellation of distinct phenomena that can accomplish a similar function of reference tracking (as [McKenzie 2012:53](#) suggests).

9 Conclusion

This paper has explored the pronominal system of Atchan as a case study in obligatory binding. One main contribution has been to argue that the long-distance obligatory binding seen here is captured via a presuppositional disjointness feature. In this way, the paper argues that at least some obligatory-binding phenomena are enforced in semantics. At the same time, however, the paper proposes a syntactic component to OBV’s orientation, since only nominals occupying the Top > Foc > Subj sequence are able to bind the distinguished variable α . It remains to be seen how the balance

between syntactic and semantic labor might play out in other obligatory-binding phenomena.

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